

Target-preserving diagnostics for household ratio indicators: a simulation study

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Abstract

Predictive models for household amounts can respond to rescaling that leaves their ratio unchanged. We specify a diagnostic that pairs household-specific common scale changes with fixed targets and observation masks, separating prediction sensitivity from estimation error. Controlled normal and mixture experiments and repeated masking of 5,219 selected Survey of Income and Program Participation (SIPP) households evaluate the share with debt-to-value above 0.8 and the mean ratio capped at 2. For a Gaussian model omitting amount covariance, an intermediate scale change reduced mean squared error (MSE) in controlled experiments, while a larger change had an uncertain effect. The largest prespecified SIPP shift increased MSE by 53.3% and 127.8%, respectively. On the original panel, empirical residual prediction improved on Gaussian prediction for both targets, but did not establish an improvement over the observed mean for the threshold target. Direct task regression had the lowest observed MSE. Finite generation increased MSE relative to exact integration, and similar average variance estimates did not ensure nominal coverage. The diagnostic identifies sensitivity whose consequences require validation against known targets. Inference assumes known, independent observation probabilities and the stated learner stability conditions.

Keywords: Monte Carlo simulation; household ratio indicators; model-assisted estimation; cross-fitting; imputation variance.

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1 Introduction

When financial amounts are incomplete, estimating a ratio indicator can require more than predicting each amount well. The share of households whose mortgage debt exceeds a fraction of residence value depends on the joint distribution of the amounts; their separate conditional means do not determine it. A fitted distribution supplies the required probability, but its usefulness must be assessed through the resulting indicator estimate. We study whether changes that preserve each household’s ratio can reveal which features of a fitted procedure matter for estimation error.

The diagnostic multiplies both amounts within a household by the same positive factor, which may differ across households. Every ratio outcome is unchanged, using the established scale-invariance principle of ratio analysis (Aitchison, 2005). We refit under the same observation masks and compare both predictions and estimates. When complete-panel targets are available for validation, this pairing isolates the effect of the scale change on estimation performance. Prediction sensitivity is not itself evidence of increased mean squared error (MSE): the correction using observed outcomes can offset a change in the predictions.

The fitting conditions make the comparison interpretable. Under a shared linear fitting operator and the stated covariance specification, jointly Gaussian log-amount prediction agrees with Gaussian log-ratio prediction and is invariant to the common shifts. Omitting the fitted amount covariance can break this equivalence. We use the independent-amount procedure to isolate covariance removal, without assuming that it is routinely used in household data analysis. The contribution is a specified diagnostic and paired evaluation of its risk consequences, rather than a new scale-invariance principle or a claim that invariance ensures accurate estimation.

The estimation ingredients are established. Model-assisted estimation uses auxiliary information to predict finite-population outcomes (Breidt and Opsomer, 2017), while fractional imputation represents missing values through weighted completions (Yang and Kim, 2016). Here fitted task means enter a common residual-corrected estimator with known observation probabilities. We compare Gaussian ratio residuals with empirical fitted residuals while holding the ratio ridge fixed. This residual averaging has precedents in conditional-expectation estimation (Müller, 2009); the efficiency results in that work do not transfer automatically to our ridge fit and pooled residual law. Direct task regression and weighted observed-data estimators provide benchmarks for deciding whether a gain over one distributional model is useful for the task itself.

The application uses the 2023 Survey of Income and Program Participation (SIPP), covering the 2022 reference year (U.S. Census Bureau, 2024b). We select 5,219 households with positive, as-reported residence values and mortgage debt. The targets are the equal-household share with debt-to-value above 0.8 and the mean ratio capped at 2. These are two bounded functions of the same ratio: one depends on threshold crossings and the other retains variation below the cap. Repeated masking of the selected records permits validation against their complete-panel means. Hypothetical common-scale changes preserve these targets and are not interpreted as observed changes in borrowing or residence values.

Controlled normal and mixture experiments separate scale variation from residual shape. A supporting experiment compares exact integration with finite generation under constant and heterogeneous observation probabilities. Fixed simulation counts have established asymptotic precedents (Pakes and Pollard, 1989), and imputation variance has been studied under other estimation and sampling laws (Robins and Wang, 2000; Hughes et al., 2016). Our comparison

concerns the specified augmented panel mean. Raw variance uses observed residuals formed from simulated predictions; split variance evaluates exact task moments and adds generation variance. Exact integration is available for both laws and tasks, providing a reference for the additional error and computation.

Each of the five experimental families has 2,000 confirmatory repetitions. Synthetic families generate new outcomes and masks, whereas SIPP masking holds the panel and shift vector fixed. Scale comparisons share targets, covariates, folds and masks within a repetition. We report paired MSE comparisons, coverage, return rates and computation times with Monte Carlo uncertainty, following Morris et al. (2019). Full conditions and numerical results are retained in the appendices and supplementary tables.

The results show why the diagnostic needs this performance comparison. An intermediate scale change reduces independent-amount MSE in the controlled experiments, while the largest change has paired MSE-ratio intervals including one. The specified SIPP shifts instead increase MSE. On the original panel, empirical residuals improve on Gaussian residuals for both targets, but their high comparison with Hájek remains uncertain; direct regression has the lowest observed MSE. Gaussian ratio’s high-task MSE exceeds Hájek’s despite its invariance. Finite generation increases MSE relative to exact integration, and close raw/split variance estimates do not ensure nominal coverage. Thus neither invariance nor improvement over a Gaussian model is sufficient evidence for choosing a method.

The variance arguments distinguish a generator fixed independently of masks from learned fits requiring the stated stability and threshold conditions. The experiments use known, independent household observation probabilities; they do not establish validity under actual SIPP nonresponse or for national population inference. With incomplete data alone, the diagnostic can expose sensitivity, but its accuracy and coverage consequences require validation targets. Section 2 defines the panel and validation design; Section 3 gives the methods and diagnostic. Sections 4 and 5 report the controlled and SIPP comparisons, and Section 6 concludes.

2 Data, targets and validation design

2.1 The SIPP validation panel

The 2023 Survey of Income and Program Participation (SIPP) public-use microdata cover the 2022 reference year (U.S. Census Bureau, 2024b). We use December records to construct a validation panel with known amount pairs. For household h , let V_h be primary-residence value (TPRVAL) and D_h the principal owed on the first three mortgages and loans against that residence (TPRLOANAMT). These variables concern non-mobile homes and are copied to household members’ records (U.S. Census Bureau, 2024a, pp. 429–432). We group records by SSUID and SHHADID, check agreement of the copied housing fields, and retain one amount pair per household.

Selection requires $V_h > 0$, $D_h > 0$, $EPRDEBT=1$, and all three status flags APRVAL, APRDEBT and APRLOAN1AMT equal to 1. The dictionary identifies these as the flags for residence value, the mortgage-or-loan indicator and the debt amount; status 1 denotes an in-universe, as-reported value (U.S. Census Bureau, 2024a, p. 8). The resulting $H = 5,219$ households form a selected validation population that excludes debt-free owners. The released amounts supply the reference outcomes; their reported status does not establish measurement accuracy or remove disclosure treatments.

Covariates X_h are age, household size, sex, race, Hispanic origin, education, region and SIPP panel identifier. Person attributes use the family-reference record, with the prespecified smallest-person-number fallback for four households. All retained households have finite age and household-size values. Covariates remain available when amounts are masked. Panel membership and the five household folds are fixed before masking and retained across comparisons.

2.2 Household ratio targets

For $r_h = D_h/V_h$, the household outcomes and their complete-panel mean are

$$q_h = \begin{pmatrix} \mathbf{1}\{r_h > 0.8\} \\ \min(r_h, 2) \end{pmatrix}, \quad \theta_F = \frac{1}{H} \sum_{h=1}^H q_h. \quad (2.1)$$

We call the two tasks *high* and *cap*. High measures the share with debt above 80% of residence value; cap averages household ratios after capping each at 2. The fixed threshold and cap provide a discontinuous indicator and a continuous bounded outcome from the same ratio. They are two tasks within one application, with equal weight for each household. The target is neither a ratio of aggregate amounts nor a survey-weighted national mean. SIPP survey and replicate weights are therefore not used.

The original panel has 528 high households and target values $\theta_{F,\text{high}} = 0.1011688063$ and $\theta_{F,\text{cap}} = 0.4754622306$. Another 43 ratios equal 0.8 and are excluded by the strict inequality. We compute q_h and the canonical log ratio $L_h = \log(r_h)$ once, preserving both across the common shifts of log amounts defined in Section 3. Synthetic outcomes are likewise fixed within a paired scale comparison. This preserves the intended target at numerical boundaries; Appendix C.1 gives the calculation details.

2.3 Observation law and paired experiments

Conditional on the complete outcomes and covariates, households are observed independently under the known law

$$R_h \stackrel{\text{ind}}{\sim} \text{Bernoulli}(p_h), \quad h = 1, \dots, H. \quad (2.2)$$

The amount pair is visible when $R_h = 1$ and hidden when $R_h = 0$. Methods receive the same X , folds, mask and probabilities, together with the visible outcomes or amount transformations they require. Hidden outcomes and complete-panel targets enter only the evaluator. Fitting and continuous-covariate standardization use visible households outside each evaluation fold.

The experiments differ in what is regenerated. E1 and E3 hold the covariates and inherited folds of a fixed 2,500-household subset constant, but generate new complete outcomes and masks in each repetition. E1 varies amount covariance and the ratio residual law; E3 studies finite generation and variance calculation. These controlled designs are specified in Section 4. E2 instead draws new masks for the same 5,219 SIPP households. At each prespecified scale, its original or shifted amounts remain fixed across repetitions. The hypothetical common shifts use one fixed vector and preserve the original ratio outcomes (Section 5). Thus the E2 comparison evaluates methods on a selected empirical distribution, whereas E1 and E3 isolate behavior under specified generating laws.

E1 and E2 use $p_h = 0.5$. E3 includes separate constant- p and covariate-dependent- p families, with the probabilities supplied to every method. In each family, methods share a mask, and

paired scale settings share both masks and ratio outcomes. Scale settings are not additional independent replications. This masking law defines the validation experiment; it is not an estimated model of actual SIPP nonresponse.

2.4 Performance and Monte Carlo uncertainty

There are 2,000 independent confirmatory repetitions per family, kept separate from the 64-repetition development runs. For repetition b and task j , error is $e_{bj} = \hat{\theta}_{bj} - \theta_{F,bj}$: the target is fixed across E2 repetitions and is recomputed for each synthetic panel. Bias, MSE and RMSE summarize these panel-relative errors. Variance calibration divides mean estimated variance by the sample variance of e_{bj} , excluding variation in the synthetic target itself.

We report Monte Carlo standard errors (MCSEs) for bias and MSE, and pointwise 95% Monte Carlo intervals for paired MSE ratios. Separately, coverage of nominal 95% estimation intervals uses all 2,000 planned repetitions; an unreturned interval counts as noncoverage. Coverage has a pointwise Wilson Monte Carlo interval. Every planned call returned both points and intervals. Return counts, interval widths and standalone times are retained with the accuracy summaries. Appendix C.5 defines these calculations and how failures would be handled; Appendix D reports every experimental setting. This separation of estimation uncertainty, Monte Carlo uncertainty and computational returns follows the reporting framework of Morris et al. (2019).

3 Estimation and target-preserving diagnostics

3.1 Methods and observed information

For a field of two-dimensional task predictions $m = (m_h)_{h=1}^H$, write

$$T_R(m) = \frac{1}{H} \left\{ \sum_{h=1}^H m_h + \sum_{h:R_h=1} \frac{q_h - m_h}{p_h} \right\}. \quad (3.1)$$

This adds an inverse-probability weighted residual correction to the predicted panel mean. It is the model-assisted form reviewed by Breidt and Opsomer (2017), applied here to the known masking law and the complete-panel target. The four fitted methods in Table 1 all use $T_R(c)$, where c_h is their fitted task mean. Their differences therefore arise from the predictions, with the target and correction held fixed. The Horvitz–Thompson (HT) baseline is $T_R(0)$ (Horvitz and Thompson, 1952). The Hájek baseline is $\hat{\theta}_{\text{Haj}} = (\sum_{R_h=1} q_h/p_h)/(\sum_{R_h=1} 1/p_h)$, the observed household mean when p_h is constant. These two baselines show whether fitting improves on procedures that use observed outcomes without a predictive model.

All learners use the covariates in Section 2.1, adding an intercept and squared standardized age. Category levels are fixed from the always-visible covariates. Each evaluation fold uses unweighted visible training rows outside that fold, with a penalty $\lambda = 0.05$ on the mean loss and an unpenalized intercept. Direct task regression fits logistic means to q_{h1} and fractional-logistic means to $q_{h2}/2$, rescaling the latter prediction by 2. Distribution methods fit log responses by ridge. Each fit requires at least 250 training households. Appendix C.2 specifies the numerical checks and return rules.

Table 1: Prediction methods and their roles in the comparison.

Method	Prediction rule	Role in the comparison
HT	No fitted prediction	Unnormalized inverse-probability baseline
Hájek	No fitted prediction	Normalized observed-outcome baseline
Direct task	Logistic high; fractional-logistic cap/2	Predict each task without a ratio distribution
Gaussian ratio	Ridge for L ; Gaussian residuals	Parametric reference for the ratio law
Empirical ratio	Same ratio ridge; empirical training residual law	Change residual shape while retaining the ratio mean fit
Independent amounts	Same ridge for both log amounts; covariance removed	Isolate sensitivity to omitting amount covariance

The four fitted methods use $T_R(c)$, $\widehat{V}_R(c)$; HT uses $T_R(0)$, $\widehat{V}_R(0)$. Hájek’s variance is given in Section 3.4. Primary distributional comparisons use exact means. E3 uses finite ratio draws at $M = 1, 20$, with the same points across variance rules. Joint Gaussian amounts and Gaussian ratio are equivalent under Proposition 3.1’s shared fits.

3.2 The common-scale diagnostic

Set $z_{hV} = \log(V_h/10^5)$ and $z_{hD} = \log(D_h/10^5)$. Adding the same s_h to both log amounts multiplies both original amounts by e^{s_h} , preserving every household’s ratio and both task outcomes. The shift may differ between households. This uses the established scale invariance of ratios (Aitchison, 2005, Section 1.4); the diagnostic asks whether the fitted task predictions also remain unchanged.

The procedure is as follows.

1. Choose shifts before comparing results; fix panel membership, covariates, folds and observation probabilities.
2. Shift visible log amounts, retaining observed q_h and canonical L_h . Keep shifts out of the covariates and verify target preservation in the validation evaluator.
3. Refit at each scale with the same mask, training rules and penalties. Couple finite predictions using the same generation uniforms.
4. Record changes in task predictions, estimates and variances, retaining return failures as in Section 2.4.
5. With known panel targets, compare errors and coverage over repeated masks using paired Monte Carlo uncertainty. Without this reference, sensitivity alone cannot establish lower MSE or valid coverage.

The following calculation explains what the diagnostic isolates for the fitted Gaussian procedures.

For evaluation fold k , let B_k be the visible training design with n_k rows and let b_{kh} be the corresponding design vector for household h . The ridge coefficient map is

$$T_k = (B_k^\top B_k / n_k + \lambda P)^{-1} B_k^\top / n_k, \quad P = \text{diag}(0, 1, \dots, 1). \quad (3.2)$$

Apply this same map to both log-amount responses. Write e_V, e_D for their training residual vectors and set $\widehat{\Sigma}_k = (e_V, e_D)^\top (e_V, e_D) / n_k + \epsilon_g I_2$, where $\epsilon_g = 10^{-8}$. A full Gaussian amount

model induces log-ratio mean $\hat{\mu}_h$ and variance $v_{J,k}$; removing its off-diagonal covariance instead gives $v_{I,k}$:

$$\begin{aligned}\hat{\mu}_h &= b_{kh}^\top T_k(z_D - z_V) = b_{kh}^\top T_k L, \\ v_{J,k} &= \overline{(e_D - e_V)^2} + 2\epsilon_g, & v_{I,k} &= \overline{e_D^2} + \overline{e_V^2} + 2\epsilon_g.\end{aligned}\tag{3.3}$$

Bars denote averages over the visible training rows. These are algebraic identities; the implementation retains the canonical outcomes and log ratios specified in Section 2.2.

Proposition 3.1 (Fitted invariance under common shifts). *Suppose the two log-amount fits share the training rows, design, penalty and response-linear map in (3.2), with an invertible regularized Gram matrix. The design and penalty do not depend on the amounts, and ϵ_g is fixed. A common household-specific shift preserves $\hat{\mu}_h$ and $v_{J,k}$. With $e_s = s - B_k T_k s$ on the training rows,*

$$v'_{I,k} - v_{I,k} = 2\overline{e_s^2} + 2\overline{e_s(e_V + e_D)}.\tag{3.4}$$

Proof. Linearity adds $T_k s$ to both coefficient vectors, leaving their contrast unchanged. Both residual vectors acquire e_s , so their difference and its second moment are unchanged. Expanding the sum of the two squared residuals gives (3.4). $\square$

The joint-amount Gaussian and Gaussian ratio predictions therefore agree under these shared fitting conditions. In contrast, the independent-amount variance responds through (3.4), whose cross term can have either sign. A single shift shared by all households is absorbed by the unpenalized intercept; household-specific shifts provide the relevant comparison. Ratio-based and direct task methods retain their inputs and predictions. The independent-amount method isolates covariance removal; it is not presented as standard SIPP practice. Separate amount-dependent designs or penalties would require a different invariance argument. Appendix A gives the full covariance change and its relation to common-scale variation and multiple log contrasts.

For any two fitted prediction fields c, c' , the same-mask change is

$$T_R(c') - T_R(c) = H^{-1} \sum_{h=1}^H (1 - R_h/p_h)(c'_h - c_h).\tag{3.5}$$

This identity holds for fitted fields on the realized mask. The resulting change in squared error has no fixed sign. Invariance is therefore a property to diagnose, while estimation accuracy must be compared against the panel target. The direct and observed-outcome benchmarks also prevent an improvement over one distributional model from being read as an improvement over every method. Appendix A.3 derives the corresponding risk difference for fixed prediction fields and explains the restriction to that setting.

3.3 Exact and simulated task predictions

Let $g(\ell) = (\mathbf{1}\{\ell > a\}, \exp\{\min(\ell, b)\})^\top$, with $a = \log 0.8$ and $b = \log 2$. Gaussian ratio uses $L_h^* \sim N(\hat{\mu}_h, v_k)$, where $\hat{\mu}_h$ is the ratio ridge prediction and $v_k = \overline{e_L^2} + 2\epsilon_g$ uses its training residuals. For a Gaussian log ratio with mean μ and variance v , the exact task mean $c = \mathbb{E}g(L^*)$

is

$$\begin{aligned} c_1 &= \Phi\{(\mu - a)/\sqrt{v}\}, \\ c_2 &= e^{\mu+v/2}\Phi\{(b - \mu - v)/\sqrt{v}\} + 2\Phi\{(\mu - b)/\sqrt{v}\}. \end{aligned} \quad (3.6)$$

Independent amounts uses the same formula with variance $v_{I,k}$.

Empirical ratio keeps the ratio ridge and draws its residual uniformly from the n_k visible training residuals. Its exact mean is

$$c_h = \frac{1}{n_k} \sum_{i \in \mathcal{T}_k} g(\hat{\mu}_h + e_{ki}), \quad (3.7)$$

where $\mathcal{T}_k$ is the visible training set. Residual averaging is established (Müller, 2009); the present comparison does not inherit that paper's efficiency result. The pooled residual law is a working distribution, not an assumption that every household has this true conditional error law. Residuals are neither smoothed nor recentered. Sorting and cumulative sums evaluate (3.7) without constructing all household-residual pairs. The implementation uses the strict comparison $e_{ki} > a - \hat{\mu}_h$ in both exact and simulated empirical predictions.

For either fitted ratio law, let $C_h = \text{Cov}\{g(L_h^*)\}$, conditional on that fitted law. Conditional draws are independent across households and draw indices, giving $m_h^* = M^{-1} \sum_{l=1}^M g(L_{hl}^*)$ for every household, including visible households. Its point estimate is $T_R(m^*)$, while exact integration uses $T_R(c)$. Fixed simulation counts have a long history in simulated estimation (Pakes and Pollard, 1989). Here $M = 1, 20$ measure the additional simulation error and computation when exact integration is available.

3.4 Variance calculation and scope

For the augmented methods use the uncentered observed residual covariance

$$\hat{V}_R(m) = H^{-2} \sum_{h:R_h=1} \frac{1-p_h}{p_h^2} (q_h - m_h)(q_h - m_h)^\top. \quad (3.8)$$

No $H/(H-1)$ multiplier is applied. In the constant- p Hájek comparisons, conditional on $n_R = \sum_h R_h \geq 2$, use

$$\hat{V}_{\text{Haj}} = \frac{1 - n_R/H}{n_R(n_R - 1)} \sum_{h:R_h=1} (q_h - \bar{q}_R)(q_h - \bar{q}_R)^\top, \quad (3.9)$$

where $\bar{q}_R$ is the observed mean. Intervals are $\hat{\theta}_j \pm 1.959964\sqrt{\hat{V}_{jj}}$, without truncation to the outcome bounds. Invalid fits or variances are recorded as failures; the fully observed Hájek case permits a zero-width interval.

The *raw* finite- M rule evaluates $\hat{V}_R(m^*)$ from the generated predictions and observed residuals, without evaluating exact task moments. First consider a generator fixed independently of all masks, with draws independent across households and draw indices and independent of the masks. Put $w_h = (1 - p_h)/p_h$. Expectation over both masks and draws gives

$$\text{Cov}_{R,*}\{T_R(m^*)\} = \mathbb{E}_{R,*}\hat{V}_R(m^*) = H^{-2} \sum_h w_h \left\{ (q_h - c_h)(q_h - c_h)^\top + C_h/M \right\}. \quad (3.10)$$

Here $T_R(m^*) - \theta_F = H^{-1} \sum_h (R_h/p_h - 1)(q_h - m_h^*)$ is a sum of independent mean-zero terms. Appendix B.1 gives the derivation. The identity specializes independent-inclusion residual variance estimation to this panel target and generation law.

The *split* rule uses the same finite- M point and draws, but evaluates exact task means and covariances for its variance calculation:

$$\hat{V}_{\text{split}} = \hat{V}_R(c) + V_{\text{MC}}, \quad V_{\text{MC}} = \frac{1}{H^2 M} \sum_h (1 - R_h/p_h)^2 C_h. \quad (3.11)$$

For a fixed mask and fitted rule, V_{MC} is covariance over the generation draws alone. Under the fixed-generator law, raw and split have equal expectations but can differ for realized draws. E3 also uses the diagnostic rules *omit*, $\hat{V}_R(c)$, and *double*, $\hat{V}_R(m^*) + V_{\text{MC}}$. They respectively omit or count generation variance twice in expectation under that law. All four rules accompany the same finite- M point estimate. These calculations concern the specified augmentation and draw scheme, rather than general multiple-imputation combining rules (Robins and Wang, 2000; Hughes et al., 2016).

The actual learners depend on observed masks and outcomes, so the exact fixed-generator identity does not directly apply. For their limit, Theorem B.1 considers a sequence of complete panels, a fixed number of folds, fixed M , and probabilities bounded away from zero and one. Each fit excludes its entire evaluation fold and does not use the generation uniforms. Let $\hat{f}_h(U)$ be its fitted single-draw task map and $f_h^0(U)$ a deterministic reference, both uniformly bounded. Couple them by common uniforms independent across households and draws and of the masks. The required task stability is

$$H^{-1} \sum_h \mathbb{E}_U \|\hat{f}_h(U) - f_h^0(U)\|^2 \rightarrow_P 0. \quad (3.12)$$

The expectation integrates a fresh uniform conditional on the fitted rule. References are proof devices, not inputs supplied to a learner. Define c_h^0, C_h^0 from f_h^0 and suppose $\Omega_H = H^{-1} \sum_h w_h \{(q_h - c_h^0)(q_h - c_h^0)^\top + C_h^0/M\}$ converges to a deterministic Ω . Also require vanishing failure probability and numerical errors negligible after $\sqrt{H}$ scaling for the point estimate and H scaling for its variance. Then

$$\sqrt{H}\{T_R(m^*) - \theta_F\} \Rightarrow N_2(0, \Omega), \quad H\hat{V}_R(m^*) \rightarrow_P \Omega, \quad H\{\hat{V}_{\text{split}} - \hat{V}_R(m^*)\} \rightarrow_P 0. \quad (3.13)$$

The proof replaces fitted fields fold by fold while retaining the evaluation masks' Bernoulli law. Boundedness and (3.12) also give the moment convergence needed for the split limit (Appendix B.3). Positive diagonal entries of Ω allow studentization. Exact integration and direct task prediction use the corresponding prediction stability condition with no generation-variance term. These limits identify the central limit theorem covariance; convergence of actual scaled finite-sample MSE needs additional tail control.

Appendix B.4 explains the learner requirements. Unweighted visible fitting has a p_h -weighted reference when observation probabilities vary. The empirical high task additionally requires control of threshold crossings; small log-scale errors alone are insufficient at an atom. The continuous cap task has no such discontinuity. Cross-fitting alone does not establish exact finite-sample covariance, and successful fits or finite boundary diagnostics do not verify the sequence assumptions. Coverage and MSE are therefore measured in the declared experiments,

including conditions unfavorable to a method.

4 Controlled simulation experiments

4.1 Prespecified design

The main experiment (E1) asks whether a method’s response to a target-preserving scale change also changes its estimation error. A second comparison within E1 holds the ratio fit fixed and varies its residual law. The supporting experiment (E3) measures the cost of finite generation relative to exact integration and checks variance calibration.

Both experiments fix the covariates and five-fold assignments of a 2,500-household subset of the SIPP panel. Let $z_{a,h}$ and $z_{n,h}$ denote age and household size standardized over this subset for outcome generation. Learners instead standardize on their visible training rows (Section 2.3). In each E1 repetition, generate

$$L_h = -1.1 + 0.35z_{a,h} - 0.15z_{n,h} + 0.7e_h,$$

$$e_h \sim \begin{cases} N(0, 1), & \text{normal structure,} \\ 0.8N(-0.4, 0.36) + 0.2N(1.6, 0.36), & \text{mixture structure.} \end{cases} \quad (4.1)$$

The second normal parameter is the variance. Both error distributions have mean zero and variance one; the mixture changes shape while retaining these moments. Errors are independent across households. With independent $Z_h \sim N(0, 1)$, set

$$A_h(s) = 0.7\sqrt{s}Z_h, \quad s \in \{0, 0.25, 1\},$$

$$z_{hV}(s) = A_h(s) - L_h/2, \quad z_{hD}(s) = A_h(s) + L_h/2. \quad (4.2)$$

Changing s changes both log amounts while leaving L_h and the two task values unchanged. Here s is the common-scale-to-log-ratio residual variance ratio. Ignoring amount covariance gives a log-ratio residual variance $1/2 + 2s$ times the true variance: $1/2, 1, 5/2$ at the three settings. At $s = 0.25$ the amount errors are uncorrelated; under mixture errors this does not imply independence. These identities describe the generating mechanism. Fitted ridge variances also reflect shrinkage, sampling variation and the fixed variance floor, so need not equal these values exactly.

E1 compares all six methods at $p_h = 0.5$, with exact task integration where applicable. Within each repetition the three scales share L, q, X , folds, Z and the mask. Their differences therefore measure the effect of changing the amounts under the same target and observed households. The two E1 structures and the two E3 probability designs each have 2,000 independent confirmatory repetitions, with fresh outcomes and masks and independent streams between families. Settings were fixed after separate 64-repetition development runs. Paired scale settings are not additional independent repetitions.

Errors are measured against each generated panel’s complete-data target. These outcome structures use SIPP covariates but are not fitted SIPP outcome distributions. All attempted estimates and intervals returned; coverage uses the planned denominator of 2,000. The fixed sample size does not assess the asymptotic learner conditions. Appendix C gives the implementation and Monte Carlo calculations, and Appendix D gives the full numerical comparisons.

4.2 Scale sensitivity and estimation error

The scale experiment separates prediction sensitivity from estimation performance. Five methods give identical estimates and standard errors across the three scales (Table 2); their duplicate rows are displayed once. The full joint Gaussian amount calculation also agrees with Gaussian ratio to numerical precision under the shared fitting operator. In contrast, the independent-amount method’s fitted task means change substantially (Figure 1, first two columns). These prediction means do not measure the bias of the augmented estimator. Its observed-data correction can offset changes in the fitted means.

Increasing s from zero to 0.25 reduces independent-amount MSE by about 4.2% for high and 2.6–2.8% for cap in both structures. For example, the normal-structure MSE ratios are 0.9580 [0.9434, 0.9729] for high and 0.9720 [0.9613, 0.9829] for cap. All four paired intervals for this change lie below one. Increasing s from zero to one gives ratios of 1.0135 [0.9768, 1.0515] and 1.0263 [0.9920, 1.0619] under normal errors; the corresponding mixture ratios are 0.9853 and 0.9987, also with intervals including one. Brackets here and in the following MSE comparisons are paired pointwise 95% Monte Carlo intervals.

Thus sensitivity to a target-preserving transformation can accompany improved MSE or a difference that remains uncertain. The intermediate setting removes the amount covariance ignored by the comparison method, which helps interpret its improvement. It does not provide a rule for choosing a transformation to improve estimation. Across E1, the largest absolute estimated bias is about 0.00011, compared with bias MCSEs of 0.00015–0.00030; the observed MSE differences mainly reflect dispersion. The risk comparison, rather than prediction sensitivity alone, determines the practical consequence in these experiments.

4.3 Residual shape and the direct-regression comparison

Holding the ratio ridge fixed isolates the choice of Gaussian or empirical residual integration. Under normal errors, the empirical-to-Gaussian MSE ratios are 1.0012 [0.9996, 1.0028] for high and 1.0000 [0.9996, 1.0005] for cap. Under mixture errors, the corresponding high ratio is 0.9836 [0.9740, 0.9932]. The cap ratio is 0.99735 [0.99474, 0.99997]: its upper endpoint lies just below one and the reduction is only about 0.27%. The high-task reduction is about 1.6%, while the cap-task reduction is small even in this mixture. Neither establishes a general advantage for empirical residual integration.

Direct task regression places these differences in context. Relative to Hájek, its MSE ratios are 0.882 [0.862, 0.902] for high and 0.783 [0.757, 0.809] for cap under normal errors; under mixture errors they are 0.954 [0.939, 0.969] and 0.847 [0.825, 0.871]. Both ratio methods also improve on Hájek in these structures (Table D.2). Their proximity to direct regression shows that fitting a distribution is not needed to obtain most of these gains. E1 therefore supports a modest, shape-dependent benefit from empirical residuals and a strong direct-regression benchmark. Their performance on the original SIPP outcomes is assessed separately in Section 5.

4.4 Finite generation: additional MSE and variance calibration

E3 uses the normal structure at $s = 0.25$ with either $p_h = 0.5$ or $p_h = 0.2 + 0.6 \expit(0.8z_{a,h})$. Gaussian and empirical ratio laws use exact integration or $M = 1, 20$ draws. The four variance rules share each finite- M point estimate and its draws; $M = 1$ uses the first draw of $M = 20$.

Table 2: Controlled scale and residual-distribution experiments.

Method	s	High		Cap	
		RMSE (MCSE)	Coverage [MC interval]	RMSE (MCSE)	Coverage [MC interval]
<i>Normal structure</i>					
HT	All	0.007756	0.9500	0.012274	0.9480
		(0.000120)	[0.9396, 0.9587]	(0.000186)	[0.9374, 0.9569]
Hájek	All	0.007109	0.9490	0.007842	0.9560
		(0.000108)	[0.9385, 0.9578]	(0.000115)	[0.9461, 0.9641]
Direct task	All	0.006676	0.9475	0.006937	0.9525
		(0.000102)	[0.9368, 0.9564]	(0.000105)	[0.9423, 0.9610]
Gaussian ratio	All	0.006667	0.9495	0.006934	0.9505
		(0.000102)	[0.9390, 0.9583]	(0.000106)	[0.9401, 0.9592]
Empirical ratio	All	0.006671	0.9490	0.006934	0.9520
		(0.000102)	[0.9385, 0.9578]	(0.000106)	[0.9417, 0.9605]
Independent amounts	0	0.006811	0.9515	0.007032	0.9520
		(0.000104)	[0.9412, 0.9601]	(0.000107)	[0.9417, 0.9605]
Independent amounts	0.25	0.006667	0.9495	0.006933	0.9510
		(0.000102)	[0.9390, 0.9583]	(0.000106)	[0.9406, 0.9596]
Independent amounts	1	0.006857	0.9490	0.007124	0.9495
		(0.000107)	[0.9385, 0.9578]	(0.000108)	[0.9390, 0.9583]
<i>Mixture structure</i>					
HT	All	0.007847	0.9435	0.013270	0.9415
		(0.000125)	[0.9325, 0.9528]	(0.000207)	[0.9303, 0.9510]
Hájek	All	0.007144	0.9470	0.009025	0.9465
		(0.000115)	[0.9363, 0.9560]	(0.000140)	[0.9358, 0.9555]
Direct task	All	0.006977	0.9475	0.008309	0.9445
		(0.000112)	[0.9368, 0.9564]	(0.000132)	[0.9336, 0.9537]
Gaussian ratio	All	0.007013	0.9475	0.008297	0.9440
		(0.000111)	[0.9368, 0.9564]	(0.000132)	[0.9330, 0.9533]
Empirical ratio	All	0.006955	0.9495	0.008286	0.9435
		(0.000111)	[0.9390, 0.9583]	(0.000131)	[0.9325, 0.9528]
Independent amounts	0	0.007165	0.9425	0.008407	0.9445
		(0.000114)	[0.9314, 0.9519]	(0.000134)	[0.9336, 0.9537]
Independent amounts	0.25	0.007012	0.9475	0.008296	0.9440
		(0.000111)	[0.9368, 0.9564]	(0.000132)	[0.9330, 0.9533]
Independent amounts	1	0.007112	0.9520	0.008402	0.9435
		(0.000111)	[0.9417, 0.9605]	(0.000132)	[0.9325, 0.9528]

Each structure has 2,000 independent outer repetitions, $H = 2,500$ and $p_h = 0.5$. The three scales share household targets and masks. “All” denotes identical estimates and variances across scales, displayed using the scale-zero rows. RMSE MCSE is $\text{MCSE}(\text{MSE})/(2 \text{RMSE})$. Coverage is return-and-cover divided by 2,000; every attempted interval returned. Brackets give pointwise 95% Wilson Monte Carlo intervals for coverage, not intervals for the household targets. They are not adjusted for multiple comparisons. Tables D.1 and D.2 give scale-specific calibration and paired MSE contrasts.

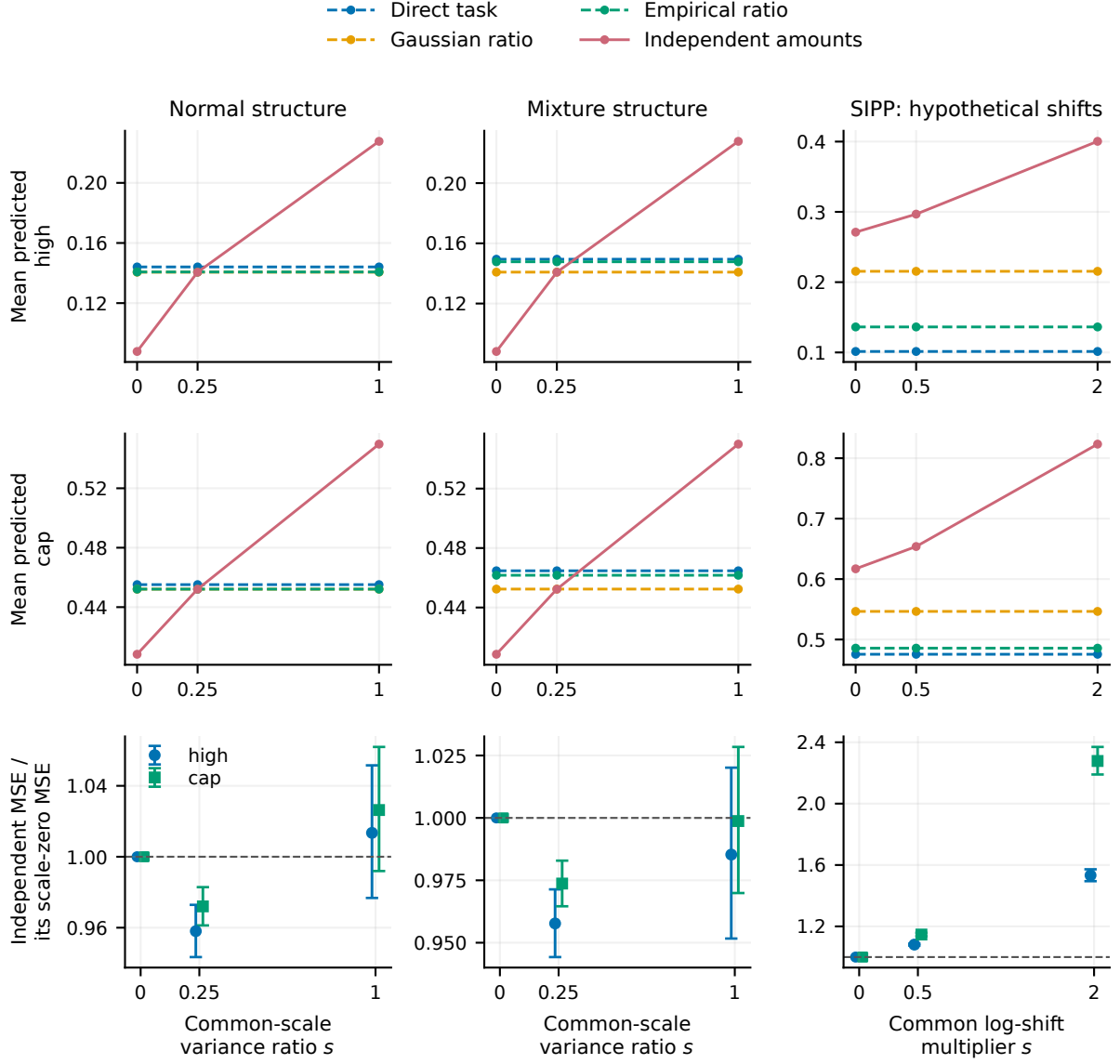


Figure 1: Target-preserving common-scale changes. Columns show the normal structure, mixture structure and SIPP panel with hypothetical shifts. The first two rows are replicate-average fitted task predictions, not augmented estimates or coverage; coincident curves can overlap. These descriptive curves have no Monte Carlo intervals. The bottom row compares independent-amount MSE with its own scale-zero MSE; bars are paired pointwise 95% Monte Carlo intervals from 2,000 repetitions, and the scale-zero ratio is identically one. Ratio-based and direct procedures are invariant. In E1, s is a residual variance ratio. For SIPP, it multiplies a fixed household-specific log shift scaled by the original standard deviation of L ; that application is discussed in Section 5. Targets and masks are paired within each family. Axes differ between columns, and connecting lines only join the three investigated settings.

This design evaluates the additional simulation error and the handling of its variance, with exact integration as the reference.

Across both probability designs, laws and tasks, finite-to-exact MSE ratios range from 1.88 to 2.07 at $M = 1$ and 1.045 to 1.056 at $M = 20$ (Figure 2). All displayed paired intervals lie above one. Twenty draws reduce the additional MSE to about 4.5–5.6%, but exact integration still has the lowest observed MSE. Table 3 reports the associated coverage, interval widths and computation times.

Raw and split variances are close on average: their ratio of means ranges from 0.9986 to 1.0028 across the 16 probability-design, law, M and task combinations. Paired coverage differences (raw minus split) range from -0.0010 to 0.0025 . For exact, raw and split procedures, mean estimated variance divided by empirical error variance ranges from 0.953 to 1.043. This agreement does not establish the fixed-generator identity for learned fits or guarantee nominal coverage.

The heterogeneous-probability empirical fit at $M = 1$ illustrates the remaining limitation. Coverage for high is 0.9415 [0.9303, 0.9510] with raw variance and 0.9390 [0.9276, 0.9487] with split variance. The latter pointwise Monte Carlo interval excludes 0.95. Their paired difference is 0.0025 with MCSE 0.00150, insufficient to establish that raw has better calibration. Results are retained without retuning. As implementation controls, omitting generation variance gives $M = 1$ coverage of 0.8245–0.8475; adding it to the raw variance (double) gives 0.9800–0.9835 (Table D.4). These controls reveal the effect of variance accounting; they are not competing estimators with different point estimates.

Standalone timings include input validation, a fresh fit and both tasks’ inference, using one BLAS thread and independently randomized method order. File I/O, raw-amount conversion and evaluator diagnostics are excluded (Appendix C.6). At $M = 1$, raw calls take 9.2–10.7 ms and split calls 12.8–17.0 ms; exact integration takes 9.2–11.4 ms. Empirical $M = 1$ raw is slightly faster than exact integration in these measurements, but has much higher MSE. At $M = 20$, exact integration is both faster and more accurate for both laws and probability designs. Raw variance avoids calculating exact generator moments when predictions are simulated. Where those moments are readily available, the present results favor exact integration; the timing rankings apply to this implementation.

Table 3: Exact integration and finite generation under two known masking laws.

Law	M /rule	High coverage [MC interval]	Cap coverage [MC interval]	Mean width high / cap	Time, ms (MCSE)
<i>Constant $p_h = 0.5$</i>					
Gaussian	Exact	0.9470	0.9470	0.02590	9.20
		[0.9363, 0.9560]	[0.9363, 0.9560]	/ 0.02705	(0.05)
Gaussian	1/raw	0.9535	0.9505	0.03639	9.22
		[0.9434, 0.9619]	[0.9401, 0.9592]	/ 0.03789	(0.07)
Gaussian	1/split	0.9520	0.9500	0.03643	12.8
		[0.9417, 0.9605]	[0.9396, 0.9587]	/ 0.03791	(0.07)
Gaussian	20/raw	0.9475	0.9430	0.02652	11.79
		[0.9368, 0.9564]	[0.9320, 0.9523]	/ 0.02770	(0.07)
Gaussian	20/split	0.9485	0.9420	0.02653	15.38
		[0.9379, 0.9574]	[0.9309, 0.9514]	/ 0.02770	(0.08)
Empirical	Exact	0.9480	0.9470	0.02590	10.55
		[0.9374, 0.9569]	[0.9363, 0.9560]	/ 0.02705	(0.05)
Empirical	1/raw	0.9520	0.9490	0.03639	9.81
		[0.9417, 0.9605]	[0.9385, 0.9578]	/ 0.03788	(0.05)
Empirical	1/split	0.9525	0.9490	0.03643	15.76
		[0.9423, 0.9610]	[0.9385, 0.9578]	/ 0.03790	(0.09)
Empirical	20/raw	0.9455	0.9430	0.02653	11.94
		[0.9347, 0.9546]	[0.9320, 0.9523]	/ 0.02770	(0.06)
Empirical	20/split	0.9465	0.9420	0.02653	17.81
		[0.9358, 0.9555]	[0.9309, 0.9514]	/ 0.02770	(0.09)
<i>Heterogeneous known probabilities</i>					
Gaussian	Exact	0.9495	0.9500	0.02313	9.99
		[0.9390, 0.9583]	[0.9396, 0.9587]	/ 0.02478	(0.05)
Gaussian	1/raw	0.9445	0.9440	0.03272	9.91
		[0.9336, 0.9537]	[0.9330, 0.9533]	/ 0.03487	(0.05)
Gaussian	1/split	0.9425	0.9430	0.03268	13.71
		[0.9314, 0.9519]	[0.9320, 0.9523]	/ 0.03486	(0.06)
Gaussian	20/raw	0.9495	0.9455	0.02370	12.93
		[0.9390, 0.9583]	[0.9347, 0.9546]	/ 0.02537	(0.08)
Gaussian	20/split	0.9490	0.9460	0.02370	16.74
		[0.9385, 0.9578]	[0.9352, 0.9551]	/ 0.02537	(0.08)
Empirical	Exact	0.9495	0.9500	0.02314	11.38
		[0.9390, 0.9583]	[0.9396, 0.9587]	/ 0.02478	(0.06)
Empirical	1/raw	0.9415	0.9430	0.03271	10.71
		[0.9303, 0.9510]	[0.9320, 0.9523]	/ 0.03485	(0.06)
Empirical	1/split	0.9390	0.9420	0.03268	16.98
		[0.9276, 0.9487]	[0.9309, 0.9514]	/ 0.03485	(0.07)
Empirical	20/raw	0.9480	0.9470	0.02370	13.00
		[0.9374, 0.9569]	[0.9363, 0.9560]	/ 0.02537	(0.07)
Empirical	20/split	0.9480	0.9475	0.02371	19.37
		[0.9374, 0.9569]	[0.9368, 0.9564]	/ 0.02537	(0.09)

There are 2,000 independent repetitions per probability design, with $H = 2,500$, normal outcomes and $s = 0.25$. Heterogeneous probabilities are $p_h = 0.2 + 0.6 \expit(0.8z_{a,h})$. “Exact” uses integrated task means; raw and split use the same finite- M point estimate and draws. Coverage intervals are pointwise 95% Wilson Monte Carlo intervals, with all 2,000 attempts in the denominator. All point estimates and intervals were returned. Widths are means of nominal 95% estimation-interval widths, given as high/cap pairs. Time is for a standalone two-task call, including fitting and inference; Appendix C.6 specifies exclusions. Tables D.4–D.5 give variance/error-variance ratios and all four variance-rule comparisons. Agreement between raw and split does not establish nominal coverage.

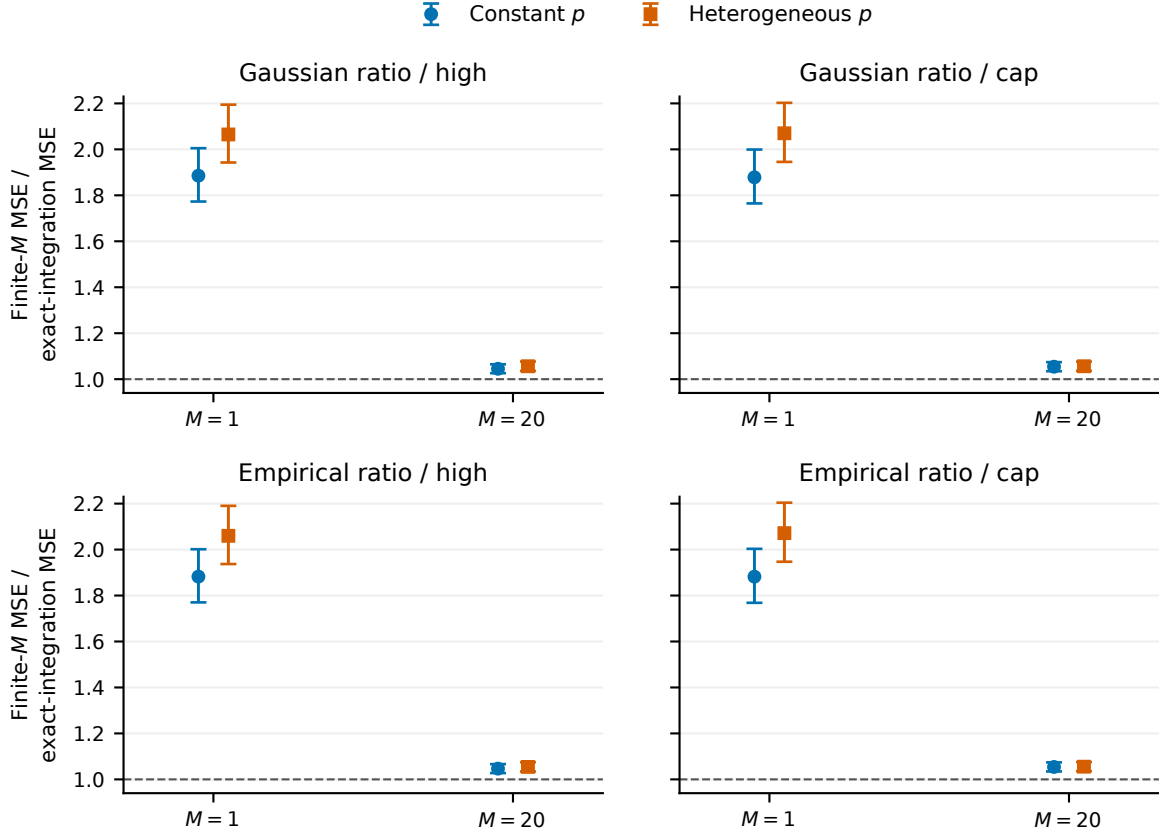


Figure 2: Finite-generation MSE relative to exact integration for the same fitted ratio method. Rows distinguish Gaussian and empirical residual laws; columns distinguish high and cap. Colors and symbols identify the two known probability designs. Bars are paired pointwise 95% Monte Carlo intervals over 2,000 repetitions per design. Within each repetition, $M = 1$ is a prefix of $M = 20$. Raw, split, omission and double-counting variance rules share the plotted point estimates and are not duplicated. Exact integration is available for both tasks; finite generation has higher empirical MSE in every displayed condition. Table 3 reports coverage, width and computation time.

5 SIPP application

5.1 Original-panel comparisons

The original-panel experiment (E2) evaluates adjustment and residual choice on the 5,219 positive, as-reported pairs in Section 2.1. Their high and cap targets, 0.1012 and 0.4755, remain fixed under 2,000 independent masks, each revealing households independently with probability 0.5. All six methods share masks, covariates and folds; distribution methods use exact integration. Table 4 reports errors relative to the fixed targets. Figure 3 compares the fitted methods with Hájek, here the observed household mean.

Direct task regression has the lowest observed RMSE for both targets among the six procedures. Its MSE relative to Hájek is 0.9544 [0.9412, 0.9678] for high and 0.8688 [0.8444, 0.8938] for cap. Brackets in the MSE comparisons are paired pointwise 95% Monte Carlo intervals. The reductions in MSE are approximately 4.56% and 13.12%, respectively. The high improvement is modest in absolute units: RMSE changes from about 0.419 to 0.410 percentage points. Direct regression provides a useful benchmark, with a small absolute gain for the high proportion.

Empirical ratio improves on Gaussian ratio for both targets. The paired MSE ratios are 0.9034 [0.8805, 0.9270] for high and 0.9196 [0.9000, 0.9396] for cap. Since the ratio ridge is shared, this comparison isolates the residual-distribution choice within the specified procedures. The comparison with the observed mean gives a different answer: empirical-to-Hájek MSE is 1.0301 [0.9921, 1.0695] for high and 0.9164 [0.8712, 0.9640] for cap. Thus empirical residuals improve the cap estimate relative to Hájek, while the high comparison remains uncertain.

For high, the Gaussian-to-Hájek MSE ratio is 1.1401 [1.0958, 1.1862], despite Gaussian ratio’s invariance to common scale. Its cap ratio is 0.9965 [0.9456, 1.0501], with no clear improvement. The independent-amount method has the largest observed RMSE for high. For cap it also performs worse than Hájek and the other three fitted methods, while HT has the largest RMSE overall. Neither residual flexibility nor invariance alone establishes an advantage over the observed mean.

Before the observed-data correction, the high predictions averaged over households and masks are 0.2154 for Gaussian ratio and 0.1363 for empirical ratio, compared with the panel target of 0.1012 (Table D.6). Empirical residuals bring the average closer to the target, but this does not measure household-level prediction error. The augmented estimator corrects most of the discrepancy; its bias is a separate quantity in Table 4. The empirical method still uses a common residual shape within each fold.

Original-panel coverage ranges from 0.9435 to 0.9580. All twelve pointwise Wilson intervals contain 0.95, consistent with nominal coverage at the available Monte Carlo precision. Several mean errors are two to three MCSE above zero, but shared masks prevent treating these comparisons as independent confirmations of bias. Table 4 retains bias and MCSE; this experiment establishes neither exact unbiasedness nor simultaneous validity.

Direct regression’s lower observed MSE comes with higher computation in this implementation. Standalone two-task calls average 67.1 ms for direct regression, 19.0 ms for empirical ratio, 16.8 ms for Gaussian ratio and 2.8 ms for Hájek. Each includes a fresh fit; these are not timings for prediction from an already fitted model.

Table 4: Original SIPP panel under repeated household masking.

Method	Bias (MCSE)	RMSE (MCSE)	Coverage [MC interval]	Mean width	Time, ms (MCSE)
<i>High</i>					
HT	0.000200 (0.000098)	0.004391 (0.000071)	0.9500 [0.9396, 0.9587]	0.01727	3.10 (0.02)
Hájek	0.000228 (0.000094)	0.004193 (0.000067)	0.9455 [0.9347, 0.9546]	0.01638	2.81 (0.02)
Direct task	0.000235 (0.000091)	0.004096 (0.000065)	0.9485 [0.9379, 0.9574]	0.01597	67.09 (0.24)
Gaussian ratio	0.000294 (0.000100)	0.004477 (0.000070)	0.9445 [0.9336, 0.9537]	0.01724	16.84 (0.08)
Empirical ratio	0.000272 (0.000095)	0.004255 (0.000066)	0.9435 [0.9325, 0.9528]	0.01647	19.00 (0.08)
Independent amounts	0.000308 (0.000107)	0.004813 (0.000075)	0.9445 [0.9336, 0.9537]	0.01851	17.82 (0.08)
<i>Cap</i>					
HT	-0.000002 (0.000168)	0.007501 (0.000122)	0.9460 [0.9352, 0.9551]	0.02953	3.10 (0.02)
Hájek	0.000125 (0.000081)	0.003626 (0.000059)	0.9550 [0.9450, 0.9632]	0.01438	2.81 (0.02)
Direct task	0.000152 (0.000076)	0.003380 (0.000053)	0.9580 [0.9483, 0.9659]	0.01331	67.09 (0.24)
Gaussian ratio	0.000201 (0.000081)	0.003619 (0.000057)	0.9440 [0.9330, 0.9533]	0.01400	16.84 (0.08)
Empirical ratio	0.000180 (0.000078)	0.003471 (0.000055)	0.9525 [0.9423, 0.9610]	0.01352	19.00 (0.08)
Independent amounts	0.000221 (0.000090)	0.004013 (0.000063)	0.9465 [0.9358, 0.9555]	0.01543	17.82 (0.08)

The fixed panel contains 5,219 positive, as-reported household amount pairs. Targets are high = 0.1011688063 and cap = 0.4754622306. There are 2,000 paired masks with $p_h = 0.5$. All values are in proportion or ratio units, except time in ms. Coverage is return-and-cover divided by 2,000; every point estimate and interval was returned. Brackets are pointwise 95% Wilson Monte Carlo intervals for coverage. RMSE MCSE uses the delta method. Times are standalone calls producing both targets, so the repeated task entries must not be added. Calls include validation, fitting, predictions, covariance and interval return, with the exclusions specified in Appendix C.6. Some mean errors are two to three MCSE above zero. These are fixed-panel validation results, not national estimates.

5.2 Hypothetical common-scale changes

For the scale diagnostic, draw independent standard normal u_h once before confirmation and hold the vector fixed across masks. Using the original-panel mean and standard deviation, set

$$\begin{aligned} \bar{L} &= H^{-1} \sum_h L_h, & \sigma_L &= \left\{ H^{-1} \sum_h (L_h - \bar{L})^2 \right\}^{1/2}, \\ (z_{hV}^{(s)}, z_{hD}^{(s)}) &= (z_{hV}, z_{hD}) + s\sigma_L u_h(1, 1), & s &\in \{0, 0.5, 2\}. \end{aligned} \quad (5.1)$$

Here σ_L is approximately 0.9833 and is fixed before masking; s is a log-shift multiplier, distinct from the E1 residual variance ratio. Original q_h, L_h , covariates, folds, masks and probabilities are retained. The panel defines the shift size and targets; learners receive only their permitted inputs (Appendix C.1). These hypothetical amount changes are not observed economic changes. Monte Carlo uncertainty is conditional on the single fixed shift vector and panel.

Figure 1, third column, shows that the independent-amount mean high prediction rises from 0.2711 at $s = 0$ to 0.4004 at $s = 2$, while every household's high outcome remains fixed. The mean cap prediction similarly rises from 0.6170 to 0.8233 without a change in any capped-ratio outcome. The corresponding MSE ratios relative to its own original-panel performance are 1.0812 [1.0749, 1.0876] for high and 1.1468 [1.1357, 1.1580] for cap at $s = 0.5$. At $s = 2$ they are 1.5329 [1.4947, 1.5720] and 2.2781 [2.1900, 2.3697]. Thus the larger shift increases MSE by approximately 53.3% and 127.8% relative to the same independent-amount method at $s = 0$, not relative to Hájek. Ratio-based and direct task predictions, estimates and variances remain unchanged; HT and Hájek are unchanged as well.

Here a change confined to the common amount scale increases estimation error. Under the controlled E1 design, however, the intermediate change improved independent-amount MSE (Section 4.2). The direction and size of the risk change therefore require validation; they do not follow from noninvariance. Table D.2 gives all paired scale contrasts, and Table D.1 retains the scale-specific bias and calibration results.

5.3 Practical interpretation

The original-panel results support evaluating a fitted distribution against both direct regression and the observed mean. A gain over its Gaussian counterpart can disappear against these task-level benchmarks, as the empirical high comparison illustrates. The observed rankings concern the two tasks and masking law studied here.

The 43 original ratios equal to 0.8 retain their strict-threshold classifications under every shift. The fitted empirical generator's exact boundary mass, averaged over households and then over masks, is approximately 2.31×10^{-7} . Raw-data atoms and predictive boundary mass are different quantities. A small measured mass on this finite panel does not establish the threshold stability condition in Section 3.4. Preserving the original task values prevents numerical reclassification from being mistaken for a method's sensitivity.

With incomplete data, common-scale changes can expose sensitivity in predictions and estimates. Assessing its MSE and coverage consequences requires validation targets; passing the diagnostic alone cannot certify accuracy or interval validity. These known- p results for selected positive, as-reported records with equal weights do not establish validity under actual SIPP nonresponse or support national population estimates.

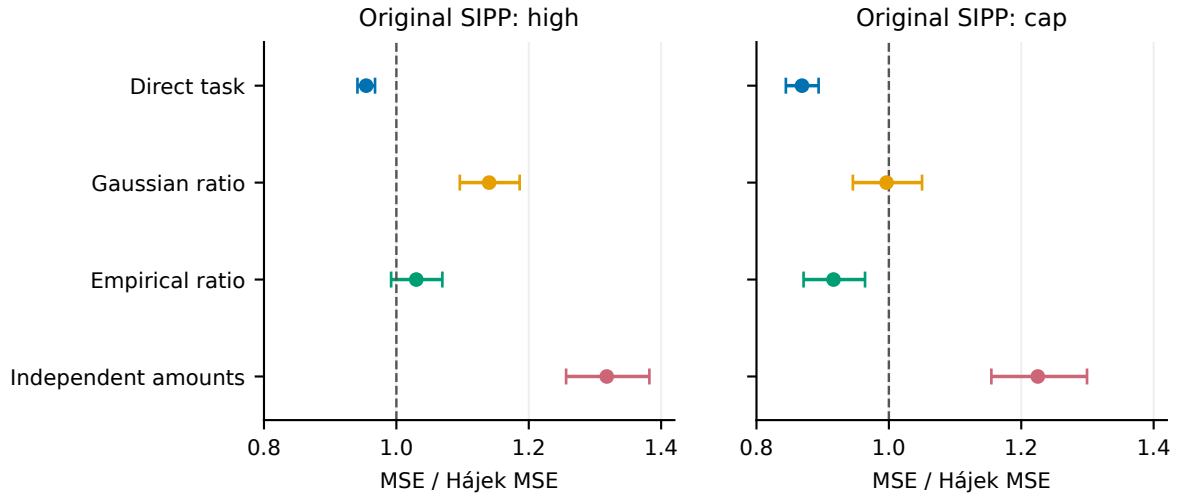


Figure 3: MSE relative to Hájek on the original SIPP panel. Points and bars are paired MSE ratios and pointwise 95% Monte Carlo intervals from the same 2,000 masks. Values below one indicate lower empirical MSE. Both panels use the same horizontal scale. All four fitted methods and both tasks are shown; HT remains in Table 4. Empirical ratio improves on Gaussian ratio for both tasks, but its high comparison with Hájek does not establish an improvement. These comparisons concern experimental masking of the fixed panel.

6 Conclusion

The study evaluates ratio estimators by changing household amount scales while preserving their targets and observation masks. This paired comparison separates prediction sensitivity from its consequences for estimation. Under the shared fitting and covariance conditions, joint Gaussian log-amount prediction agrees with Gaussian log-ratio prediction. Removing amount covariance can break that equivalence. The diagnostic uses this established ratio invariance to identify sensitivity, then measures its importance against known panel targets.

The risk consequences differ across designs. An intermediate common-scale change reduced independent-amount MSE in E1, while the largest change had paired MSE-ratio intervals that included one. The specified SIPP shifts increased MSE for both tasks, conditional on the fixed panel and shift vector. Noninvariance therefore does not determine the direction of a risk change. Conversely, Gaussian ratio was invariant but had higher high-task MSE than Hájek on the original SIPP panel.

These results support comparing a fitted distribution with direct regression and the observed mean. Direct task regression had the lowest observed RMSE for both SIPP targets, with greater computation in this implementation. Empirical residuals improved on Gaussian residuals for both targets, but improved on Hájek only for cap; the high comparison remained uncertain. A gain over one residual model is thus insufficient evidence for choosing distribution-based estimation.

In E3, finite generation increased MSE relative to exact integration for both fitted laws and tasks. These results favor exact integration when the required task moments are readily available. When using simulated predictions, raw-residual variance avoids exact moment calculation and was faster than split accounting, with similar mean estimated variances. Their agreement did not ensure nominal coverage: the empirical high task with one draw under heterogeneous probabilities retained a split-variance coverage shortfall. The paired comparison did not establish that raw accounting resolved it.

The exact covariance identity requires a generator independent of the masks and the specified independent draws. The extension to learned fits requires the stated sampling and learner stability assumptions, including threshold control for the empirical high task. Cross-fitting and successful finite-panel diagnostics do not verify those conditions. The SIPP results concern selected positive, as-reported records with equal weights and known masking probabilities; they do not establish validity under actual nonresponse or for national population inference.

In applications, the comparison should preserve task values and available information, refit under the same masks, and record changes in both predictions and estimates. Where validation targets are available, their MSE and coverage consequences can then be assessed alongside task-level benchmarks. With incomplete data alone, the diagnostic can expose sensitivity but cannot certify accuracy or interval validity.

A Algebra and target-preserving perturbations

This appendix supplies the covariance and risk calculations used in Section 3.2. The fitting identities hold for each realized set of visible training rows satisfying the stated conditions. Expectations over masks enter only in Appendix A.3, where the prediction fields are fixed. The identities are over the real numbers; the implementation preserves the canonical observed ratios and outcomes defined in Section 2.2 when changing the log amounts.

A.1 Shared fits and common-scale variation

For fold k , collect the visible training log amounts in $Z_k = (z_V, z_D)$, an $n_k \times 2$ matrix, and the common shifts in $s_k \in \mathbb{R}^{n_k}$. Retain the design B_k and map T_k from (3.2). Let $\mathbf{1}_2 = (1, 1)^\top$ and $d = (-1, 1)^\top$. The transformation is $Z'_k = Z_k + s_k \mathbf{1}_2^\top$. Because $\mathbf{1}_2^\top d = 0$,

$$b_{kh}^\top T_k Z'_k d = b_{kh}^\top T_k Z_k d = b_{kh}^\top T_k L. \quad (\text{A.1})$$

Thus the fitted log-ratio mean is unchanged and equals the direct ridge fit to L . This uses the same response-linear map for both amounts; separate response-dependent penalties or designs need not satisfy (A.1).

Define the residual matrix and residual shift by

$$E_k = (I_{n_k} - B_k T_k) Z_k, \quad u_k = (I_{n_k} - B_k T_k) s_k. \quad (\text{A.2})$$

Then $E'_k = E_k + u_k \mathbf{1}_2^\top$. Writing $\gamma_k = E_k^\top u_k / n_k$ and $\tau_k = u_k^\top u_k / n_k$, direct expansion of the residual covariance gives

$$\widehat{\Sigma}'_k - \widehat{\Sigma}_k = \gamma_k \mathbf{1}_2^\top + \mathbf{1}_2 \gamma_k^\top + \tau_k \mathbf{1}_2 \mathbf{1}_2^\top. \quad (\text{A.3})$$

The fixed $\epsilon_g I_2$ cancels from this difference. Premultiplying by $d^\top$ and postmultiplying by d gives zero, so the joint Gaussian log-ratio variance is unchanged. Taking the trace instead gives

$$v'_{I,k} - v_{I,k} = 2\tau_k + 2\mathbf{1}_2^\top \gamma_k = 2\overline{e_s^2} + 2\overline{e_s(e_V + e_D)}, \quad (\text{A.4})$$

where $e_s = u_k$ and the bars average over the training rows. The mixed term is part of the fitted change and cannot generally be discarded.

The unchanged Gaussian mean and variance imply the same distribution for the log ratio and the same expectations of both bounded tasks. The ratio and direct task methods also retain their observed inputs. Their deterministic predictions, augmented estimates and observed residual variances are therefore unchanged. For a finite- M ratio method, this last statement holds with the same ratio-scale random numbers. Independent draws in the two runs would introduce simulation differences even when the fitted laws agree. These claims concern fits that are defined and use the same numerical rules.

The common-scale component can be isolated by setting $e_A = (e_V + e_D)/2$ and $e_L = e_D - e_V$. Expanding the residual squares yields

$$v_{J,k} = \overline{e_L^2} + 2\epsilon_g, \quad v_{I,k} = 2\overline{e_A^2} + \frac{1}{2}\overline{e_L^2} + 2\epsilon_g. \quad (\text{A.5})$$

For square-integrable random variables A, L , let $z_V = A - L/2$ and $z_D = A + L/2$. The

corresponding population identities are

$$\begin{aligned}\text{Cov}(z_V, z_D) &= \text{Var}(A) - \frac{1}{4} \text{Var}(L), \\ \text{Var}(z_V) + \text{Var}(z_D) &= 2 \text{Var}(A) + \frac{1}{2} \text{Var}(L).\end{aligned}\tag{A.6}$$

The two cross terms involving $\text{Cov}(A, L)$ cancel; these identities do not require independence of A and L . Applied to the structural residuals, with $\text{Var}(L) > 0$ and $\rho = \text{Var}(A)/\text{Var}(L)$, the variance ratio after removing covariance is $1/2 + 2\rho$. The levels $\rho = 0, 1/4, 1$ therefore give $1/2, 1, 5/2$. Those population ratios do not assert exact fitted ratios in (A.5), where ridge shrinkage, finite training samples and the variance floor remain.

A shift constant across households has $u_k = 0$: the unpenalized intercept reproduces a constant response exactly. Household-specific shifts need not have this property. Their trace change can have either sign; for example, $u_k = -e_A$ gives $v'_{I,k} - v_{I,k} = -2\overline{e_A^2}$, whereas $u_k = e_A$ gives $6\overline{e_A^2}$. These choices follow from taking the training shift to be minus or plus $(z_V + z_D)/2$. They are algebraic examples of the possible signs, not additional experiment settings.

A.2 Multiple log contrasts

The same calculation extends to several contrasts. Let $Z \in \mathbb{R}^d$, $C \in \mathbb{R}^{r \times d}$ and $G \in \mathbb{R}^{d \times \ell}$ be such that $CG = 0$. Then $C(Z + G\eta) = CZ$ for every $\eta \in \mathbb{R}^\ell$; every task that depends only on CZ is preserved. Scale invariance and log contrasts are established principles (Aitchison, 2005, Sections 1.4 and 2.5). The following covariance calculation states how diagonalization can affect this property.

For a covariance matrix Σ and $t \geq 0$, consider the specified path $\Sigma_t = \Sigma + tGG^\top$. Its full contrast covariance satisfies

$$C\Sigma_t C^\top = C\Sigma C^\top.\tag{A.7}$$

Let $\mathcal{D}(S) = \text{diag}(S_{11}, \dots, S_{dd})$ and let a_j denote column j of C . Since diagonal element j of $GG^\top$ is $\|G_{j\cdot}\|^2$,

$$C\{\mathcal{D}(\Sigma_t) - \mathcal{D}(\Sigma)\}C^\top = t \sum_{j=1}^d \|G_{j\cdot}\|^2 a_j a_j^\top.\tag{A.8}$$

Every summand is positive semidefinite. For $t > 0$, the sum vanishes exactly when $a_j = 0$ at every nonzero row $G_{j\cdot}$: taking its trace gives a sum of nonnegative terms $t\|G_{j\cdot}\|^2\|a_j\|^2$. At $t = 0$ it vanishes without this restriction. For $G = \mathbf{1}_d$ and $C\mathbf{1}_d = 0$, the change is $tCC^\top$.

The path $\Sigma + tGG^\top$ describes, for example, adding a centered component $G\eta$ uncorrelated with Z and having $\text{Cov}(\eta) = tI_\ell$. Its nonnegative diagonalized change does not determine the sign of a general fitted change: (A.3) also includes mixed terms. Neither covariance calculation establishes an ordering of task estimation risk.

A.3 Prediction changes and fixed-field risk

Condition on the complete panel and let R_h be independent Bernoulli(p_h) variables with $0 < p_h < 1$. Put $\alpha_h = 1 - R_h/p_h$ and $w_h = (1 - p_h)/p_h$. For any two prediction fields c, c' , including realized fitted fields, let $\delta_h = c'_h - c_h$. Subtracting the augmented estimates gives

$$\Delta_R := T_R(c') - T_R(c) = H^{-1} \sum_h \alpha_h \delta_h.\tag{A.9}$$

For task j , the corresponding realized squared-error change is

$$\{T_R(c')_j - \theta_{F,j}\}^2 - \{T_R(c)_j - \theta_{F,j}\}^2 = 2\{T_R(c)_j - \theta_{F,j}\}\Delta_{R,j} + \Delta_{R,j}^2. \quad (\text{A.10})$$

The first term can be negative. Thus a nonzero prediction or estimate change alone does not determine whether the error increases.

Now take c, c' to be fixed independently of all masks and set $e_h = q_h - c_h$. Since $T_R(c) - \theta_F = H^{-1} \sum_h (R_h/p_h - 1)e_h$, independent mean-zero masking terms give

$$\mathbb{E}_R T_R(c) = \theta_F, \quad \text{Cov}_R\{T_R(c)\} = H^{-2} \sum_h w_h e_h e_h^\top. \quad (\text{A.11})$$

Replacing e_h by $e_h - \delta_h$ and expanding yields

$$\begin{aligned} & \text{Cov}_R\{T_R(c')\} - \text{Cov}_R\{T_R(c)\} \\ &= H^{-2} \sum_h w_h \{\delta_h \delta_h^\top - e_h \delta_h^\top - \delta_h e_h^\top\}. \end{aligned} \quad (\text{A.12})$$

Both estimators are unbiased under this fixed-field law, so their coordinate MSE differences are the diagonal entries of (A.12). For any fixed task contrast $\xi \in \mathbb{R}^2$,

$$\text{MSE}_R\{\xi^\top T_R(c')\} - \text{MSE}_R\{\xi^\top T_R(c)\} = H^{-2} \sum_h w_h \{(\xi^\top \delta_h)^2 - 2(\xi^\top e_h)(\xi^\top \delta_h)\}, \quad (\text{A.13})$$

where MSE is relative to $\xi^\top \theta_F$. There is no universal sign.

Equations (A.9) and (A.10) remain algebraic identities for learned fields. Equations (A.11)–(A.13) require fields fixed independently of the masks; conditioning on all fitted fields does not generally preserve the independent masking law. Predictions learned outside their evaluation folds can still be dependent across folds. Their risk in Sections 4 and 5 is assessed by repeated fitting under the declared experiments.

B Variance calculations and learner conditions

Throughout this appendix, condition on a complete panel and its fold partition. The target is its two-dimensional mean θ_F , and the only observed outcomes available to a fitted procedure are those with $R_h = 1$. Write $\alpha_h = 1 - R_h/p_h$ and $w_h = (1 - p_h)/p_h$ as in Appendix A.3. All covariance matrices below refer to this panel target and the stated masking and generation law. The arguments specialize the residual correction in model-assisted estimation (Breidt and Opsomer, 2017). They do not provide a general variance rule for other imputation or sampling procedures.

B.1 A fixed generator and the source of simulation variance

First take the generator to be fixed independently of all masks. For household h , let Q_{hl}^* , $l = 1, \dots, M$, be identically distributed bounded task vectors with mean c_h and covariance C_h . Draws are independent across households and replicates and independent of the independent Bernoulli(p_h) masks. Set $m_h^* = M^{-1} \sum_l Q_{hl}^*$ for every household, including those with $R_h = 1$. We require $0 < p_h < 1$ and finite $M \geq 1$.

The centered estimator is

$$T_R(m^*) - \theta_F = H^{-1} \sum_h (R_h/p_h - 1)(q_h - m_h^*). \quad (\text{B.1})$$

Each summand has mean zero. Independence across households eliminates cross-household covariance terms, while $\mathbb{E}_R(R_h/p_h - 1)^2 = w_h$ and

$$\mathbb{E}_*\{(q_h - m_h^*)(q_h - m_h^*)^\top\} = (q_h - c_h)(q_h - c_h)^\top + C_h/M. \quad (\text{B.2})$$

It follows that

$$\begin{aligned} \mathbb{E}_{R,*} T_R(m^*) &= \theta_F, \\ \text{Cov}_{R,*}\{T_R(m^*)\} &= H^{-2} \sum_h w_h \{(q_h - c_h)(q_h - c_h)^\top + C_h/M\} = \mathbb{E}_{R,*} \hat{V}_R(m^*). \end{aligned} \quad (\text{B.3})$$

For the last equality, substitute (3.8) and use $\mathbb{E}_R\{R_h(1 - p_h)/p_h^2\} = w_h$ together with (B.2). This proves (3.10) without a degrees-of-freedom multiplier. Conditioning first on all preassigned draws gives the same result by fixed-field residual variance calculation and averaging.

For a fixed mask, write $T_R(m^*) = T_R(c) + H^{-1} \sum_h \alpha_h(m_h^* - c_h)$. Hence

$$\mathbb{E}_*\{T_R(m^*) \mid R\} = T_R(c), \quad \text{Cov}_*\{T_R(m^*) \mid R\} = H^{-2} M^{-1} \sum_h \alpha_h^2 C_h = V_{\text{MC}}(R). \quad (\text{B.4})$$

Since $\mathbb{E}_R \alpha_h^2 = w_h$, define $S_M = H^{-2} M^{-1} \sum_h w_h C_h$. The two variance calculations satisfy

$$\mathbb{E}_{R,*} \hat{V}_R(m^*) = \mathbb{E}_R\{\hat{V}_R(c) + V_{\text{MC}}(R)\} = H^{-2} \sum_h w_h (q_h - c_h)(q_h - c_h)^\top + S_M. \quad (\text{B.5})$$

They can differ for a realized mask and draw set. Under this fixed-rule law, using $\hat{V}_R(c)$ alone omits S_M in expectation, and adding V_{MC} to $\hat{V}_R(m^*)$ includes it twice. These expectation statements do not imply exact finite-sample coverage.

The conditional path formula (B.4) also holds after fixing an observed mask and its fitted generator, provided the subsequent draws are independent across households. Averaging over refitted generators requires an additional argument, given below. Similarly, a superpopulation mean has complete-record variation that is absent from error relative to the same panel's mean. Changing that target would require a different variance calculation.

B.2 A limit for predictions learned outside the evaluation fold

Consider a sequence of deterministic complete panels indexed by H ; the index H on household quantities is suppressed. Partition each panel into a fixed number K of folds $\mathcal{F}_k$. Masks are independent, with $\varepsilon_p \leq p_h \leq 1 - \varepsilon_p$ for a constant $\varepsilon_p > 0$. For $h \in \mathcal{F}_k$, the fitted task map $\hat{f}_h$ is measurable with respect to the masks and observed outcomes outside $\mathcal{F}_k$, the fixed covariates and any auxiliary training randomization W . The latter is independent of all masks and generation uniforms. Thus the fit excludes every mask in its evaluation fold and does not use the uniforms that will generate its predictions.

Let U_{hl} be independent $\text{Uniform}(0, 1)$ variables, independent of all masks and W . Fix $M \geq 1$. For deterministic reference maps f_h^0 , set

$$m_h^* = M^{-1} \sum_{l=1}^M \hat{f}_h(U_{hl}), \quad m_h^0 = M^{-1} \sum_{l=1}^M f_h^0(U_{hl}). \quad (\text{B.6})$$

References may depend on H , the complete panel, the household and fold, but on no masks or auxiliary randomness. They are proof devices, not inputs to the implemented estimators. Suppose q_h , $\hat{f}_h$ and f_h^0 are uniformly bounded, and assume the integrated task condition from (3.12):

$$\mathcal{E}_H := H^{-1} \sum_h \mathbb{E}_U \|\hat{f}_h(U) - f_h^0(U)\|^2 \longrightarrow_P 0. \quad (\text{B.7})$$

Here $\mathbb{E}_U$ integrates a fresh uniform conditional on the fitted rule. Put $c_h^0 = \mathbb{E}_U f_h^0(U)$, $C_h^0 = \text{Cov}_U\{f_h^0(U)\}$ and assume

$$\Omega_H = H^{-1} \sum_h w_h \{(q_h - c_h^0)(q_h - c_h^0)^\top + C_h^0/M\} \longrightarrow \Omega \quad (\text{B.8})$$

for a deterministic positive semidefinite matrix Ω .

Theorem B.1 (Conditional limit for learned task predictions). *Under the preceding assumptions,*

$$\sqrt{H}\{T_R(m^*) - \theta_F\} \Rightarrow N_2(0, \Omega), \quad H\hat{V}_R(m^*) \longrightarrow_P \Omega. \quad (\text{B.9})$$

Any fixed task contrast with positive limiting variance is asymptotically standard normal after studentization.

Proof. First replace the fitted fields by their references. The summands $\zeta_h = (R_h/p_h - 1)(q_h - m_h^0)$ are independent, centered and uniformly bounded. Their covariance sum divided by H is Ω_H . Lindeberg's condition therefore holds for each fixed contrast; the scalar central limit theorem and the Cramér–Wold argument give the first limit for $T_R(m^0)$. Each entry of $H\hat{V}_R(m^0)$ is an average of independent bounded variables with expectation equal to the corresponding entry of Ω_H . Its variance is $O(H^{-1})$, giving the second limit for the reference.

Let $\Delta_h = m_h^* - m_h^0$ and $D_H = H^{-1} \sum_h \|\Delta_h\|^2$. Jensen's inequality gives $\mathbb{E}(D_H \mid R, W) \leq \mathcal{E}_H$. Conditional Markov's inequality, with the probability bound truncated at one, implies $D_H = o_P(1)$ from (B.7). For each fold condition on $\mathcal{G}_k = \sigma(R_{-\mathcal{F}_k}, W, (U_{hl})_{h,l})$. The Δ_h for $h \in \mathcal{F}_k$ are then fixed and the evaluation masks retain their independent Bernoulli laws. With $Z_k = H^{-1/2} \sum_{h \in \mathcal{F}_k} \alpha_h \Delta_h$,

$$\mathbb{E}(Z_k \mid \mathcal{G}_k) = 0, \quad \mathbb{E}(\|Z_k\|^2 \mid \mathcal{G}_k) = H^{-1} \sum_{h \in \mathcal{F}_k} w_h \|\Delta_h\|^2 \leq \kappa D_H. \quad (\text{B.10})$$

The middle expression, which is $\mathcal{G}_k$ -measurable, tends to zero in probability. Conditional Chebyshev's inequality, again truncated at one before taking expectations, gives $Z_k = o_P(1)$. Consequently $\sqrt{H}\{T_R(m^*) - T_R(m^0)\} = \sum_{k=1}^K Z_k = o_P(1)$. This step uses fixed K and requires no independence between folds.

For the variance, let $r_h^0 = q_h - m_h^0$ and $\hat{r}_h = q_h - m_h^*$. The outer-product identity and Cauchy–Schwarz inequality give, for a constant κ independent of H ,

$$\begin{aligned} & \|H\{\hat{V}_R(m^*) - \hat{V}_R(m^0)\}\|_F \\ & \leq \kappa D_H^{1/2} \left\{ H^{-1} \sum_h (\|\hat{r}_h\| + \|r_h^0\|)^2 \right\}^{1/2} = o_P(1). \end{aligned} \quad (\text{B.11})$$

The residuals are uniformly bounded. Combining the reference limits with these two replacements proves (B.9). Slutsky's theorem gives studentization for positive limiting contrast variance. $\square$

For exact integration, use $m_h^* = \hat{c}_h = \mathbb{E}_U \hat{f}_h(U)$ and $m_h^0 = c_h^0$, and omit C_h^0/M from (B.8). The same proof requires average squared prediction differences to vanish. This also covers bounded direct task predictions with a deterministic reference. Consistency for the true conditional task mean is not required by this replacement argument.

For an implementation that can fail, the proof uses a bounded, outside-fold measurable extension on failure events, whose probability must tend to zero. Such an extension is only an analytic device: the experiments retain failed calls in their declared denominators. Numerical discrepancies must be $o_P(H^{-1/2})$ for the point estimate and $o_P(H^{-1})$ for its variance. A successful fit or fixed solver tolerance does not verify those sequence conditions. Finally, (B.9) identifies the limiting normal covariance. Convergence of actual scaled finite-sample MSE or variance additionally needs suitable uniform integrability of the scaled errors.

B.3 Why split and raw variances have the same limit

Write $\hat{c}_h = \mathbb{E}_U \hat{f}_h(U)$ and $\hat{C}_h = \text{Cov}_U\{\hat{f}_h(U)\}$. Uniform boundedness and (B.7) give

$$H^{-1} \sum_h \|\hat{c}_h - c_h^0\|^2 \leq \mathcal{E}_H = o_P(1), \quad H^{-1} \sum_h \|\hat{C}_h - C_h^0\|_F \leq \kappa \mathcal{E}_H^{1/2} = o_P(1). \quad (\text{B.12})$$

The first inequality is Jensen's. For the second, expand covariance as a second moment minus the outer product of the mean, use boundedness to bound each outer-product difference by a constant times the task difference, then apply Cauchy–Schwarz over households and the uniform.

The fitted split calculation is

$$\widehat{V}_{\text{split}} = \widehat{V}_R(\widehat{c}) + H^{-2}M^{-1} \sum_h \alpha_h^2 \widehat{C}_h. \quad (\text{B.13})$$

Bounded weights and (B.12) allow $\widehat{c}_h, \widehat{C}_h$ to be replaced by c_h^0, C_h^0 after multiplication by H , with an $o_P(1)$ remainder. The resulting reference is an average of independent bounded mask terms. Since $\mathbb{E}_R \alpha_h^2 = w_h$, its expectation is Ω_H and its entrywise variance is $O(H^{-1})$. Therefore

$$H\widehat{V}_{\text{split}} \rightarrow_P \Omega, \quad H\{\widehat{V}_{\text{split}} - \widehat{V}_R(m^*)\} \rightarrow_P 0. \quad (\text{B.14})$$

The integrated condition supplies the fitted moment replacements; agreement for one realized set of generated values would not supply them. The raw implementation requires no evaluation of those exact moments. These limits do not make raw and split identical in a finite sample, or establish that either attains nominal coverage in every experimental setting.

B.4 Conditions for the fitted predictors

The reference under unequal observation probabilities. Let $\mathcal{I}_k$ contain all households outside fold k , and put $\pi_k = \sum_{i \in \mathcal{I}_k} p_i$. The visible training set is $\mathcal{T}_k = \{i \in \mathcal{I}_k : R_i = 1\}$. Because training uses unweighted visible mean loss, its reference uses full-training moments weighted by p_i/π_k . Denote the design vector standardized using those weights by b_{ki}^0 . With the same penalty matrix P ,

$$\beta_k^0 = \left\{ \pi_k^{-1} \sum_{i \in \mathcal{I}_k} p_i b_{ki}^0 b_{ki}^{0\top} + \lambda P \right\}^{-1} \pi_k^{-1} \sum_{i \in \mathcal{I}_k} p_i b_{ki}^0 L_i. \quad (\text{B.15})$$

Set $\mu_{kh}^0 = b_{kh}^{0\top} \beta_k^0$ and $r_{ki}^0 = L_i - b_{ki}^{0\top} \beta_k^0$. The Gaussian residual variance and empirical residual distribution have references

$$v_k^0 = \pi_k^{-1} \sum_{i \in \mathcal{I}_k} p_i (r_{ki}^0)^2 + 2\epsilon_g, \quad F_k^0 = \pi_k^{-1} \sum_{i \in \mathcal{I}_k} p_i \delta_{r_{ki}^0}, \quad (\text{B.16})$$

where δ_x is a unit point mass at x . These quantities use complete-panel values only for the proof. The feasible fits never receive hidden responses. Constant p_i gives unweighted full-training references; unequal p_i generally does not. The fixed penalty $\lambda = 0.05$ also remains in the reference.

One sufficient route to ridge mean and residual-moment stability uses fixed design dimension, uniformly bounded complete-panel covariates and log responses, training fractions bounded below, reference standardization scales bounded away from zero and bounded inverses of the reference regularized Gram matrices. Independent masking then gives convergence of the visible moments to their weighted references and $|\mathcal{T}_k|/\pi_k \rightarrow_P 1$. Continuity of standardization and matrix inversion gives coefficient convergence and average squared prediction convergence. Weaker tail assumptions may also suffice, but invertibility or finite moments on one panel do not establish any such sequence condition. Empirical residual distribution stability requires the additional condition below.

Log draws and the two bounded tasks. Let $\widehat{\ell}_h(U)$ and $\ell_h^0(U)$ be coupled fitted and reference log draws. Suppose

$$\eta_H := H^{-1} \sum_h \mathbb{E}_U |\widehat{\ell}_h(U) - \ell_h^0(U)|^2 \longrightarrow_P 0. \quad (\text{B.17})$$

The capped task $g_2(\ell) = \exp\{\min(\ell, \log 2)\}$ is globally 2-Lipschitz, so its average squared task difference is at most $4\eta_H$. For the strict threshold at $a = \log 0.8$, every $\delta > 0$ satisfies

$$|\mathbf{1}\{\widehat{\ell}_h > a\} - \mathbf{1}\{\ell_h^0 > a\}| \leq \mathbf{1}\{|\ell_h^0 - a| \leq \delta\} + \mathbf{1}\{|\widehat{\ell}_h - \ell_h^0| > \delta\}. \quad (\text{B.18})$$

The squared difference of two indicators equals their absolute difference. Averaging (B.18) and using Markov's inequality bounds its integrated mean by $H^{-1} \sum_h \Pr_U(|\ell_h^0 - a| \leq \delta) + \eta_H/\delta^2$. It therefore suffices that

$$\lim_{\delta \downarrow 0} \limsup_{H \rightarrow \infty} H^{-1} \sum_h \Pr_U(|\ell_h^0 - a| \leq \delta) = 0. \quad (\text{B.19})$$

First let H grow for fixed δ , then let δ decrease. This gives (B.7) for both tasks.

For Gaussian draws, couple through a common $Z \sim N(0, 1)$: $\widehat{\ell}_h = \widehat{\mu}_h + \sqrt{\widehat{v}_k} Z$ and $\ell_h^0 = \mu_h^0 + \sqrt{v_k^0} Z$. Average squared mean differences and squared differences of the square-root variances tending to zero give (B.17). The fixed floor $v_k^0 \geq 2\epsilon_g > 0$ uniformly bounds the reference normal densities, so their probabilities in intervals of length 2δ are at most a constant times δ . This supplies (B.19). A floor tending to zero would require a separate boundary argument.

For empirical residual draws, use generalized inverse distribution functions and the same uniform:

$$\widehat{\ell}_h(U) = \widehat{\mu}_h + \widehat{F}_k^{-1}(U), \quad \ell_h^0(U) = \mu_h^0 + (F_k^0)^{-1}(U), \quad h \in \mathcal{F}_k. \quad (\text{B.20})$$

In addition to mean prediction stability, require, for each fold,

$$\int_0^1 |\widehat{F}_k^{-1}(u) - (F_k^0)^{-1}(u)|^2 du \longrightarrow_P 0. \quad (\text{B.21})$$

The squared difference of the two log draws is bounded by twice the squared mean difference plus twice the squared quantile difference. Fixed K then gives (B.17). The threshold task still needs (B.19) or an explicit allowance for stable atoms. The cap task does not need a boundary-mass condition.

To allow stable atoms, replace $|\ell_h^0 - a| \leq \delta$ in (B.19) by $0 < |\ell_h^0 - a| \leq \delta$, and additionally require

$$H^{-1} \sum_h \Pr_U\{\ell_h^0 = a, \widehat{\ell}_h > a\} \longrightarrow_P 0. \quad (\text{B.22})$$

Split a threshold mismatch into an exact reference atom, a nonzero distance at most δ and a log-draw difference exceeding δ . The same bound proves task stability. For a reference atom exactly at a , the strict indicator is zero, so only a crossing to $\widehat{\ell}_h > a$ changes it. In contrast, $\ell_h^0 = a$ and $\widehat{\ell}_h = a + 1/H$ give vanishing squared log differences but a threshold mismatch of one for every H . Increasing M cannot repair this discrepancy in the exact task mean. Rounded observed SIPP ratios and atoms of its fitted predictive distribution are different objects; the recorded boundary diagnostics do not prove these asymptotic conditions.

Direct task predictions. For direct logistic and fractional-logistic fits, a sufficient local route uses stable design and estimating-equation moments, a finite reference solution, local nonsingularity of the true gradient derivative and negligible numerical error. Smoothness and boundedness of the rescaled logistic predictions then give the mean replacement condition. A floored working Hessian or a returned numerical solution does not by itself verify those requirements. The fractional response need not have a Bernoulli distribution for this estimating equation.

B.5 Cross-fitting does not imply exact covariance

For a scalar task with $H = 2$, let $p_1 = p_2 = 1/2$, $q_1 = q_2 = 0$ and $\hat{m}_1 = R_2$, $\hat{m}_2 = R_1$. Each bounded prediction excludes its own mask. The four equally likely masks give

$$\begin{array}{c|cccc}
 (R_1, R_2) & (0, 0) & (1, 0) & (0, 1) & (1, 1) \\
 \hline
 T_R(\hat{m}) & 0 & 1/2 & 1/2 & -1 \\
 \hat{V}_R(\hat{m}) & 0 & 0 & 0 & 1
 \end{array} \tag{B.23}$$

Thus $\mathbb{E}_R T_R(\hat{m}) = 0$, $\text{Var}_R\{T_R(\hat{m})\} = 3/8$ and $\mathbb{E}_R \hat{V}_R(\hat{m}) = 1/4$. Cross-fold dependence prevents the fixed-generator covariance identity. More generally, an always-defined bounded outside-fold prediction preserves unbiasedness by conditioning separately for each household, but that mean identity does not establish exact covariance. Conditioning on successful returns can also alter the mean. Theorem B.1 handles the covariance through replacement by a deterministic reference under its stated conditions; it does not assume the dependence away.

C Experimental design and implementation

This appendix records the design and numerical rules used for the confirmatory comparisons. The design was fixed after separate 64-repetition development runs and before the 2,000-repetition confirmatory families. This was an internal freeze of the protocol and code, not registration in a public registry. Development results are not pooled with confirmation. The description separates aims, data generation, estimands, methods and performance measures, following the reporting framework of Morris et al. (2019).

C.1 Panels, observation boundaries and comparison units

The archived SIPP panel has the selection, household records and fixed five folds described in Section 2.1. A fixed simple random subset of 2,500 of its 5,219 households supplies the covariates and inherited folds for E1 and E3. Subset indices are sorted back into the original row order. Category levels are fixed from the complete panel’s always-observed covariates; sorted levels omit their first category as the reference. The generating age and household-size standardizations use this entire fixed subset, with divisor H for their variances. Learner standardizations instead use only each fold’s visible training rows.

The normal and mixture E1 families follow (4.1)–(4.2) at $s = 0, 0.25, 1$. Each repetition draws new independent ratio errors, common-scale shocks and masks. The same errors, shocks, outcomes, covariates, folds and mask are used at all three scales within that repetition. E3 uses the normal structure at $s = 0.25$; its constant- and heterogeneous-probability families have separate panel and mask streams. The latter uses $p_h = 0.2 + 0.6 \expit(0.8z_{a,h})$ and, on the fixed design, has probability range approximately $[0.28348, 0.72166]$.

E2 holds the original SIPP outcomes fixed. Its three log-scale multipliers are 0, 0.5, 2 in (5.1). The common-shift vector is generated once as $\sigma_L Z_h$, with independent standard normal Z_h and $\sigma_L = 0.9833188012$, the original panel standard deviation of the canonical log ratio with divisor H . The realized vector is stored and reused; it is not restandardized, redrawn after seeing results, or supplied as a covariate. All three panels receive the same new Bernoulli(0.5) mask in a repetition. The shift changes the amount distribution while leaving the ratio targets and eligibility of the original households fixed.

The five families have 2,000 independent outer repetitions each. The three scale conditions in each E1 or E2 repetition are paired cases, not additional independent repetitions. Thus there are 10,000 outer repetitions and $3 \times 2,000 \times 3 + 2 \times 2,000 = 22,000$ panel cases. E1 and E2 each use six deterministic methods. An E3 case has $2\{1 + 2 \times 4\} = 18$ method configurations: two laws, exact integration, and two finite draw counts with four variance rules. The resulting 180,000 two-task calls give 360,000 scalar task records; these call and record counts do not increase the number of independent outer repetitions.

Each feasible call receives X , folds, R , the known vector p and arrays containing observed L , q and log amounts, with hidden rows marked missing. Input validation also selects visible rows before examining outcomes or amount transformations. Panel means and hidden outcomes enter only the evaluator after the call returns. The two amounts are hidden together, and no method trains on hidden targets.

For observed SIPP records, first compute $r = D/V$, then $L = \log r$ and $q = (\mathbf{1}\{r > 0.8\}, \min(r, 2))^\top$. Retain these values across scale changes and transform the log amounts

directly. Of the 43 original ratios equal to 0.8, seven would change strict classification if $\log(D/10^5) - \log(V/10^5)$ were compared with $\log 0.8$ instead. This is a floating-point boundary discrepancy, not a change in the recorded household target. Synthetic q and L are likewise defined once from the generated L . The validator checks log-contrast and capped outcome consistency to absolute tolerance 2×10^{-10} ; it does not repair a boundary discrepancy with hidden information.

C.2 Fitting and numerical return rules

Every fold uses the same intercept, standardized age, squared standardized age, standardized household size and category indicators. Continuous standard deviations use the visible training count as divisor and must exceed 10^{-8} . The minimum visible training count is 250. Fits exclude the entire evaluation fold. The mean-loss penalty is $\lambda = 0.05$, with the intercept unpenalized, and training is unweighted even when observation probabilities differ.

For ridge, the code solves (3.2) and checks that the maximum absolute gradient component is at most 10^{-8} . The two log amounts share the same design and operator. Gaussian ratio uses the mean squared training log-ratio residual plus $2\epsilon_g$, where $\epsilon_g = 10^{-8}$. Independent amounts uses the trace of the two-amount residual second-moment matrix plus $2\epsilon_g$. The empirical law uses the same ratio ridge and its sorted residual pool, without smoothing, recentering or degrees-of-freedom rescaling.

Direct task fitting applies the logistic mean-loss objective to q_1 and $q_2/2$ and rescales the second prediction by 2. Newton updates use the true logistic Hessian and an Armijo search: the initial step length is one, successive lengths are halved, and at most 31 candidates are considered. The sufficient-decrease coefficient is 10^{-4} with additive numerical slack 10^{-14} . The algorithm allows at most 100 outer iterations, stops when the gradient maximum is at most 10^{-10} , and applies the final 10^{-8} gradient certification. No sigmoid clipping or working-Hessian floor is used. An all-zero or all-one training response, singular solve, failed search or uncertified gradient is a recorded fit failure.

All task covariance matrices are checked for finite values and positive semidefiniteness. When subtraction of exact second moments produces a negative eigenvalue no smaller than -10^{-10} , the code projects only this roundoff component to zero and records the repair. A more negative eigenvalue is a failure. The raw residual covariance is computed as a weighted sum of outer products. The final interval check uses its own 10^{-12} tolerance for covariance eigenvalues and requires a positive diagonal variance for a returned Wald interval. The fully observed Hájek case alone permits a zero-width interval. Neither estimates nor interval endpoints are truncated to the target bounds, and a failed method is not replaced by another estimator.

C.3 Exact task moments and simulated predictions

Write $Q = g(\ell)$ with $a = \log 0.8$ and $b = \log 2$. For $\ell \sim N(\mu, v)$, let $s_v = \sqrt{v}$ and use the means in (3.6). The second moments needed for the split variance are

$$\begin{aligned} \mathbb{E}Q_1^2 &= c_1, \\ \mathbb{E}Q_2^2 &= e^{2\mu+2v}\Phi\{(b-\mu-2v)/s_v\} + 4\Phi\{(\mu-b)/s_v\}, \\ \mathbb{E}(Q_1Q_2) &= e^{\mu+v/2}[\Phi\{(b-\mu-v)/s_v\} - \Phi\{(a-\mu-v)/s_v\}] + 2\Phi\{(\mu-b)/s_v\}. \end{aligned} \quad (\text{C.1})$$

Then $C = \mathbb{E}(QQ^\top) - cc^\top$. Gaussian tail and interval probabilities use log-CDF calculations and stable log differences. The mean-only calculation does not evaluate second moments.

For a sorted empirical pool $e_{(1)}, \dots, e_{(n)}$, define $n_a = \#\{i : e_i \leq a - \mu\}$, $n_b = \#\{i : e_i \leq b - \mu\}$ and $S_k(t) = \sum_{i:e_i \leq t} \exp(ke_i)$, $k = 1, 2$. Exact averaging gives

$$\begin{aligned} c_1 &= 1 - n_a/n, & c_2 &= e^\mu S_1(b - \mu)/n + 2(1 - n_b/n), \\ \mathbb{E}Q_1^2 &= c_1, & \mathbb{E}Q_2^2 &= e^{2\mu} S_2(b - \mu)/n + 4(1 - n_b/n), \\ \mathbb{E}(Q_1 Q_2) &= \frac{e^\mu}{n} \{S_1(b - \mu) - S_1(a - \mu)\} + 2(1 - n_b/n). \end{aligned} \quad (\text{C.2})$$

Log cumulative sums and binary searches evaluate these expressions. For a nonempty interval whose cumulative log sums differ by less than 10^{-12} , the second-moment calculation uses a direct log-sum-exp over that interval and records the fallback. The usual sorting and query cost per fold is $O(n \log n + n_{\text{eval}} \log n)$; this fallback can add work. No dense evaluation-household by training-residual matrix is formed.

Both exact and simulated empirical paths use the same strict comparison $e_i > a - \hat{\mu}_h$. The draw index is $1 + \lfloor nU \rfloor$ in the sorted pool for $0 < U < 1$; the cap is computed as $\exp\{\min(\hat{\mu}_h + e_i, b)\}$. Gaussian draws use $\hat{\mu}_h + \sqrt{v_k} \Phi^{-1}(U)$. The implementation moves any numerical uniform endpoint to the adjacent floating-point value inside $(0, 1)$. It generates tasks for every household and averages the M task vectors, including at observed households.

In each E3 law and repetition, $M = 1$ is the first row of the replicate-major $M = 20$ uniform array. The same array is used across Gaussian and empirical laws and across raw, split, omission and double-counting variance rules. Each configuration is nevertheless fitted in a separate call. The raw finite- M call evaluates no exact task moments; split and the two diagnostic rules additionally evaluate the exact means and covariances. The code uses $M = 0$ as the flag for exact integration, corresponding to the paper's $M = \infty$ reference.

Fitted empirical boundary diagnostics average, over evaluation households, the fraction of training residuals in the closed interval $[a - \hat{\mu}_h - \delta, a - \hat{\mu}_h + \delta]$ for $\delta = 0, 10^{-12}, 10^{-8}, 10^{-4}$. The zero-width case measures exact predictive boundary mass. These diagnostics use fitted pools and visible covariates; they are distinct from the original SIPP atom count and do not verify the conditions in Appendix B.

C.4 Random streams and paired comparisons

NumPy `SeedSequence` uses master entropy (20260920, 20, 731904), followed by a stage, family, zero-based repetition and purpose identifier. Stage 1 is development and stage 2 is confirmation. Family identifiers are 10 and 11 for E1 normal and mixture, 20 for E2, and 30 and 31 for the constant and heterogeneous E3 designs. Purpose identifiers 0, 1, 2, 3 and 4 separate ratio generation, common-scale shocks, masks, prediction uniforms and method-order randomization. The latter appends the scale index. `default_rng` draws from each specified sequence.

The initial subset and SIPP shift use the separate entropy extensions (0, 0, 0, 0) and (0, 0, 0, 1) and are shared with development as fixed design inputs. Confirmation uses repetition identifiers 0 through 1999. Families have separate random streams, while comparisons within a family and repetition share the inputs specified above. Reusing an identical seed across separately evaluated configurations implements the intended coupling; it does not make their results independent.

The run stores masks, subset indices and the SIPP shift, together with case targets, attempt

identities and hashes of inputs and code. SIPP confirmation masks were checked for internal duplicates and against the 64 development masks and five earlier mask archives. These are identity checks; independence is a property of the specified generation design rather than a conclusion from absence of collisions. Paired analyses join by family, scale as appropriate, task and outer repetition. They never treat the three scales or four finite- M variance rules as independent resamples.

C.5 Error summaries, uncertainty and failures

For a fixed family, method, setting and task, let $e_b = \hat{\theta}_b - \theta_{F,b}$, $b = 1, \dots, B$, with $B = 2,000$. Bias is $\bar{e}$, MSE is $\bar{e}^2$, and RMSE is $\sqrt{\bar{e}^2}$. Their reported bias and MSE MCSEs are $\text{sd}_B(e_b)/\sqrt{B}$ and $\text{sd}_B(e_b^2)/\sqrt{B}$, where sd_B uses divisor $B - 1$. The variance-calibration ratio divides the mean reported variance by $\text{var}_B(e_b)$, not by the variance of uncentered estimates across newly generated panels.

For a paired comparison, put $u_b = e_{A,b}^2$, $v_b = e_{B,b}^2$ and $r = \bar{u}/\bar{v}$, assuming both mean squared errors are positive. The reported pointwise Monte Carlo interval is

$$\exp \left[\log r \pm z_{.975} \frac{\text{sd}_B(u_b/\bar{u} - v_b/\bar{v})}{\sqrt{B}} \right], \quad z_{.975} = 1.959963984540054. \quad (\text{C.3})$$

The paired MSE difference is $\bar{u} - \bar{v}$ with MCSE $\text{sd}_B(u_b - v_b)/\sqrt{B}$. Coverage differences use the analogous MCSE of paired cover indicators. These quantify simulation uncertainty for each stated contrast and are not simultaneous tests of general superiority across all conditions.

Let I_b be one if an interval is returned and covers its own panel target, and zero otherwise. Planned-denominator coverage is $\hat{c} = B^{-1} \sum_b I_b$, with MCSE $\sqrt{\hat{c}(1 - \hat{c})/B}$. Its pointwise Wilson interval is

$$\frac{\hat{c} + z_{.975}^2/(2B) \pm z_{.975} \sqrt{\hat{c}(1 - \hat{c})/B + z_{.975}^2/(4B^2)}}{1 + z_{.975}^2/B}. \quad (\text{C.4})$$

Point and interval return rates and failure reasons are recorded separately. Error summaries would be conditional on point return if some estimates failed; paired error comparisons would use pairs with both points returned and report their count. Mean reported variance, interval width and returned-interval coverage use returned intervals. Every planned call returned both points and intervals in confirmation, so those conditional and planned counts coincide here.

The recorded diagnostics include training counts, gradients, iterations and backtracking, design conditioning, covariance repairs, point and interval status, and prediction-array hashes. Evaluator prediction errors use hidden truth only after estimation. Joint-amount versus ratio Gaussian means and variances are checked to 2×10^{-10} . Identical finite- M points across variance rules and invariant predictions across paired scales are checked separately from the comparisons of MSE or coverage.

C.6 Timing and reproducibility records

Each standalone timer starts at canonical observed input and includes input validation, fitting, prediction or generation, covariance, standard errors, the nominal 95% Wald intervals and return status. It excludes raw-file input/output, conversion of raw amounts to canonical inputs, evaluator panel and mask construction, and post-return truth diagnostics. All methods and

scale configurations are refitted separately without a fit cache. Their execution order is a seeded random permutation within each case, independent of masks and prediction paths. Timing therefore measures the stated implementation boundary rather than only a moment formula.

The recorded environment is 64-bit Windows with Python 3.12.10, NumPy 2.5.1, SciPy 1.18.0 and pandas 2.3.3. BLAS calls use one thread. Timing means and their MCSEs summarize the 2,000 standalone calls for each configuration; they are machine-specific measurements. The 180,000 recorded call times sum to 2,666.3449 seconds, while the recorded estimation-session wall time is 3,233.4942 seconds. The two quantities have different boundaries, and neither includes subsequent export recovery or analysis.

The implementation separates feasible methods (`methods.py`), evaluator panels and streams (`study_io.py`), execution (`run_development.py`, with its explicit confirmation option) and paired summaries (`summarize.py`). The SIPP public-use source is identified in U.S. Census Bureau (2024b). Local records retain the source-panel hash, fixed inputs, masks, planned calls, case checkpoints, scalar outcomes, timing observations and the recorded software environment. A completed run is protected against overwriting. The accompanying `reproduction/` package provides the extracted panel, original estimation code, frozen confirmation inputs, scalar outputs and a separate command for raw-data panel construction. Its entry point `reproduce.py` supports a reduced replay, the full confirmation run, and regeneration of summaries and figures in new output directories. The README records the commands, environment and checks. The reproduction notes document export history and distinguish measured times from development-based projections.

D Additional comparisons and computational results

This appendix reports every prespecified experimental setting, including the comparisons that limit a claim of improvement. Each task-specific summary uses all 2,000 planned repetitions; all points and intervals were returned. A finite-panel target is used in each repetition, as specified in Appendix C.5. High denotes the share with debt-to-value above 0.8; cap denotes the mean ratio capped at 2.

The accompanying machine-readable tables retain all 180 performance summaries, 328 paired comparisons, 90 timing summaries and 16 raw-split comparisons. The filenames and their contents are listed at the end of this appendix. The performance tables combine exact duplicates: invariant methods are reported once across scales, and point estimates shared by finite- M variance rules are reported once. Coverage and timing remain separate wherever the procedures differ. The paired tables show baseline method comparisons, all nonzero shifts, finite versus exact integration, and residual-law comparisons; other contrasts remain in the full file.

D.1 Accuracy, calibration and paired comparisons

Table D.1 includes all six methods in E1 and E2. In these tables, R_V is mean estimated variance divided by the sample variance of panel-relative errors. It is not a ratio of standard errors or of uncentered estimates across synthetic panels. A dagger identifies a pointwise Wilson interval for coverage that excludes 0.95. It is a descriptive calibration flag, without adjustment across conditions. Bias, its MCSE, interval width and the Wilson endpoints are retained for every condition in `supplementary/summary.csv`.

Table D.1: E1 and E2: accuracy and calibration across all scales.

Method	Scale	RMSE $\times 10^3$		R_V		Coverage	
		High	Cap	High	Cap	High	Cap
<i>E1: normal residuals</i>							
HT	All	7.756	12.274	0.958	0.952	0.9500	0.9480
Hájek	All	7.109	7.842	0.977	0.985	0.9490	0.9560
Direct	All	6.676	6.937	0.988	0.995	0.9475	0.9525
Gaussian ratio	All	6.667	6.934	0.983	0.991	0.9495	0.9505
Empirical ratio	All	6.671	6.934	0.983	0.991	0.9490	0.9520
Independent amounts	0	6.811	7.032	0.977	0.983	0.9515	0.9520
Independent amounts	0.25	6.667	6.933	0.984	0.991	0.9495	0.9510
Independent amounts	1	6.857	7.124	0.992	1.010	0.9490	0.9495
<i>E1: mixture residuals</i>							
HT	All	7.847	13.270	0.971	0.934	0.9435	0.9415
Hájek	All	7.144	9.025	0.996	0.959	0.9470	0.9465
Direct	All	6.977	8.309	0.994	0.959	0.9475	0.9445
Gaussian ratio	All	7.013	8.297	0.994	0.958	0.9475	0.9440
Empirical ratio	All	6.955	8.286	0.996	0.960	0.9495	0.9435
Independent amounts	0	7.165	8.407	0.983	0.951	0.9425	0.9445
Independent amounts	0.25	7.012	8.296	0.994	0.959	0.9475	0.9440
Independent amounts	1	7.112	8.402	1.006	0.974	0.9520	0.9435
<i>E2: selected SIPP panel</i>							
HT	All	4.391	7.501	1.009	1.008	0.9500	0.9460
Hájek	All	4.193	3.626	0.997	1.025	0.9455	0.9550
Direct	All	4.096	3.380	0.992	1.011	0.9485	0.9580
Gaussian ratio	All	4.477	3.619	0.969	0.977	0.9445	0.9440
Empirical ratio	All	4.255	3.471	0.979	0.991	0.9435	0.9525
Independent amounts	0	4.813	4.013	0.966	0.965	0.9445	0.9465
Independent amounts	0.5	5.005	4.297	0.964	0.960	0.9475	0.9470
Independent amounts	2	5.959	6.057	0.959	0.951	0.9420	0.9430

All means $s = 0, 0.25, 1$ in E1 and $s = 0, 0.5, 2$ in E2; the five invariant methods have identical statistical summaries at these scales. Independent amounts omits the fitted amount covariance. All distributional predictions here use exact integration. Coverage uses nominal 95% Wald intervals and denominator 2,000. †: the pointwise 95% Wilson interval excludes 0.95.

Table D.2 gives paired MSE ratios with the pointwise 95% Monte Carlo intervals in equation (C.3). The baseline comparisons include HT and direct regression as well as the distributional methods. The scale comparisons show all nonzero shifts for the independent-amount procedure. For each of the other five methods, every paired scale ratio and both interval endpoints equal one exactly.

The E1 intermediate shift reduces MSE for both tasks under both residual laws, while all four E1 largest-shift intervals include one. The SIPP shifts increase MSE. Thus sensitivity alone has no fixed direction of effect on estimation accuracy. On the original SIPP panel, direct regression has the lowest observed MSE; empirical residual prediction improves on Gaussian ratio prediction but does not establish an improvement over Hájek for high. The small mixture cap difference between empirical and Gaussian prediction is retained with five-decimal interval endpoints so that rounding does not obscure its proximity to one.

Table D.2: Paired MSE ratios: baseline methods and all nonzero scale changes.

Numerator	Denominator	High: ratio [MC interval]	Cap: ratio [MC interval]
<i>E1: normal residuals, scale zero</i>			
HT	Hájek	1.1903 [1.15150, 1.23041]	2.4498 [2.30358, 2.60523]
Direct	Hájek	0.8817 [0.86164, 0.90229]	0.7826 [0.75733, 0.80877]
Gaussian ratio	Hájek	0.8794 [0.85396, 0.90560]	0.7819 [0.75205, 0.81300]
Empirical ratio	Hájek	0.8805 [0.85499, 0.90668]	0.7819 [0.75205, 0.81304]
Independent amounts	Hájek	0.9178 [0.89069, 0.94582]	0.8042 [0.77454, 0.83504]
Empirical ratio	Gaussian ratio	1.0012 [0.99957, 1.00283]	1.0000 [0.99958, 1.00047]
<i>E1: mixture residuals, scale zero</i>			
HT	Hájek	1.2068 [1.16764, 1.24717]	2.1619 [2.03586, 2.29565]
Direct	Hájek	0.9539 [0.93937, 0.96875]	0.8475 [0.82482, 0.87077]
Gaussian ratio	Hájek	0.9637 [0.93789, 0.99022]	0.8452 [0.81800, 0.87325]
Empirical ratio	Hájek	0.9479 [0.92964, 0.96645]	0.8429 [0.81658, 0.87014]
Independent amounts	Hájek	1.0060 [0.97733, 1.03544]	0.8677 [0.84021, 0.89612]
Empirical ratio	Gaussian ratio	0.9836 [0.97401, 0.99324]	0.9973 [0.99474, 0.99997]
<i>E2: selected SIPP panel, scale zero</i>			
HT	Hájek	1.0969 [1.06594, 1.12873]	4.2799 [3.95430, 4.63241]
Direct	Hájek	0.9544 [0.94122, 0.96776]	0.8688 [0.84439, 0.89384]
Gaussian ratio	Hájek	1.1401 [1.09582, 1.18624]	0.9965 [0.94564, 1.05014]
Empirical ratio	Hájek	1.0301 [0.99207, 1.06948]	0.9164 [0.87115, 0.96400]
Independent amounts	Hájek	1.3178 [1.25640, 1.38223]	1.2250 [1.15484, 1.29936]
Empirical ratio	Gaussian ratio	0.9034 [0.88049, 0.92700]	0.9196 [0.90004, 0.93958]
<i>Independent amounts: transformed / scale zero</i>			
E1 normal	Scale 0.25 / 0	0.9580 [0.94343, 0.97288]	0.9720 [0.96128, 0.98286]
E1 normal	Scale 1 / 0	1.0135 [0.97677, 1.05154]	1.0263 [0.99195, 1.06193]
E1 mixture	Scale 0.25 / 0	0.9577 [0.94425, 0.97137]	0.9737 [0.96459, 0.98288]
E1 mixture	Scale 1 / 0	0.9853 [0.95167, 1.02013]	0.9987 [0.96990, 1.02843]
E2 SIPP	Scale 0.5 / 0	1.0812 [1.07485, 1.08757]	1.1468 [1.13570, 1.15802]
E2 SIPP	Scale 2 / 0	1.5329 [1.49470, 1.57200]	2.2781 [2.19003, 2.36966]

Each contrast uses 2,000 matched repetitions. Ratios below one favor the numerator for that setting. The intervals describe Monte Carlo uncertainty and are not simultaneous evidence of superiority. The complete 328-row file also includes method comparisons at each nonzero scale, invariant scale contrasts and every finite- M rule.

D.2 Finite generation: accuracy and variance rules

Table D.3 compares finite generation with exact integration for each law and probability regime. The four variance rules share exactly the same point estimate, so their MSE ratios coincide. Exact integration has lower MSE in every finite-generation comparison. Comparing the two laws at the same M gives much smaller differences, with all displayed law-comparison intervals including one. These comparisons do not support a statistical advantage from finite simulation when the required task moments are available directly.

Table D.3: E3: point accuracy, finite-generation cost and residual-law comparisons.

Law	M	RMSE $\times 10^3$		MSE / exact [MC interval]		Cap
		High	Cap	High		
<i>E3: constant probabilities</i>						
Gaussian ratio	∞	6.664	6.916	1 (reference)		1 (reference)
Gaussian ratio	1	9.151	9.478	1.8854	[1.77294, 2.00504]	1.8783 [1.76479, 1.99921]
Gaussian ratio	20	6.813	7.100	1.0452	[1.02610, 1.06473]	1.0542 [1.03455, 1.07414]
Empirical ratio	∞	6.667	6.915	1 (reference)		1 (reference)
Empirical ratio	1	9.147	9.487	1.8823	[1.77021, 2.00144]	1.8822 [1.76838, 2.00325]
Empirical ratio	20	6.822	7.100	1.0468	[1.02754, 1.06642]	1.0541 [1.03450, 1.07400]
<i>E3: heterogeneous probabilities</i>						
Gaussian ratio	∞	5.952	6.301	1 (reference)		1 (reference)
Gaussian ratio	1	8.552	9.066	2.0648	[1.94280, 2.19450]	2.0700 [1.94542, 2.20247]
Gaussian ratio	20	6.117	6.476	1.0562	[1.03615, 1.07665]	1.0562 [1.03644, 1.07631]
Empirical ratio	∞	5.955	6.301	1 (reference)		1 (reference)
Empirical ratio	1	8.547	9.070	2.0598	[1.93705, 2.19029]	2.0716 [1.94707, 2.20405]
Empirical ratio	20	6.114	6.473	1.0543	[1.03426, 1.07464]	1.0552 [1.03555, 1.07529]
<i>Empirical / Gaussian ratio at the same generation size</i>						
Constant p	∞	–	–	1.0009	[0.99928, 1.00254]	1.0000 [0.99954, 1.00041]
Constant p	1	–	–	0.9992	[0.98651, 1.01213]	1.0020 [0.99832, 1.00569]
Constant p	20	–	–	1.0024	[0.99828, 1.00655]	0.9999 [0.99885, 1.00093]
Heterogeneous p	∞	–	–	1.0011	[0.99930, 1.00284]	1.0001 [0.99963, 1.00052]
Heterogeneous p	1	–	–	0.9986	[0.98460, 1.01286]	1.0009 [0.99706, 1.00467]
Heterogeneous p	20	–	–	0.9992	[0.99464, 1.00383]	0.9992 [0.99800, 1.00035]

$M = \infty$ denotes exact integration (encoded as $M = 0$ in the files). Upper panels use the same law and probability regime as denominator; the lower panel uses Gaussian ratio at the same M . All ratios use 2,000 pairs and pointwise 95% Monte Carlo intervals. Raw, split, omit and double have identical points within a law and M ; their distinct variance estimates and intervals appear in Table D.4.

Coverage provides a separate comparison. Table D.4 includes the diagnostic omit and double rules as well as raw and split. At $M = 1$, omitting generation variance gives coverage between 0.8245 and 0.8475, while counting it twice gives 0.9800 to 0.9835. Raw and split are much closer to nominal coverage, but do not achieve it uniformly. For empirical prediction with heterogeneous probabilities and $M = 1$, high coverage is 0.9415 for raw and 0.9390 for split; the latter's pointwise Wilson interval is [0.92765, 0.94867].

Table D.4: E3: every exact and finite-generation variance configuration.

Law	M	Rule	High		Cap	
			R_V	Coverage	R_V	Coverage
<i>E3: constant probabilities</i>						
Gaussian ratio	∞	Exact	0.984	0.9470	0.997	0.9470
Gaussian ratio	1	raw	1.030	0.9535	1.043	0.9505
Gaussian ratio	1	split	1.031	0.9520	1.043	0.9500
Gaussian ratio	1	omit	0.522	0.8410 [†]	0.531	0.8475 [†]
Gaussian ratio	1	double	1.540	0.9830 [†]	1.554	0.9830
Gaussian ratio	20	raw	0.987	0.9475	0.992	0.9430
Gaussian ratio	20	split	0.987	0.9485	0.992	0.9420
Gaussian ratio	20	omit	0.941	0.9385 [†]	0.946	0.9375 [†]
Gaussian ratio	20	double	1.033	0.9545	1.037	0.9455
Empirical ratio	∞	Exact	0.983	0.9480	0.997	0.9470
Empirical ratio	1	raw	1.031	0.9520	1.040	0.9490
Empirical ratio	1	split	1.032	0.9525	1.041	0.9490
Empirical ratio	1	omit	0.522	0.8390 [†]	0.530	0.8440 [†]
Empirical ratio	1	double	1.541	0.9835 [†]	1.551	0.9830 [†]
Empirical ratio	20	raw	0.985	0.9455	0.992	0.9430
Empirical ratio	20	split	0.985	0.9465	0.992	0.9420
Empirical ratio	20	omit	0.939	0.9395 [†]	0.946	0.9370 [†]
Empirical ratio	20	double	1.031	0.9535	1.037	0.9460
<i>E3: heterogeneous probabilities</i>						
Gaussian ratio	∞	Exact	0.986	0.9495	1.010	0.9500
Gaussian ratio	1	raw	0.956	0.9445	0.967	0.9440
Gaussian ratio	1	split	0.953	0.9425	0.966	0.9430
Gaussian ratio	1	omit	0.478	0.8260 [†]	0.488	0.8350 [†]
Gaussian ratio	1	double	1.431	0.9805 [†]	1.444	0.9820 [†]
Gaussian ratio	20	raw	0.980	0.9495	1.002	0.9455
Gaussian ratio	20	split	0.980	0.9490	1.002	0.9460
Gaussian ratio	20	omit	0.934	0.9435	0.956	0.9420
Gaussian ratio	20	double	1.026	0.9540	1.049	0.9515
Empirical ratio	∞	Exact	0.985	0.9495	1.009	0.9500
Empirical ratio	1	raw	0.956	0.9415	0.965	0.9430
Empirical ratio	1	split	0.954	0.9390 [†]	0.965	0.9420
Empirical ratio	1	omit	0.478	0.8245 [†]	0.488	0.8335 [†]
Empirical ratio	1	double	1.431	0.9800 [†]	1.442	0.9820 [†]
Empirical ratio	20	raw	0.981	0.9480	1.003	0.9470
Empirical ratio	20	split	0.981	0.9480	1.003	0.9475
Empirical ratio	20	omit	0.934	0.9435	0.956	0.9415
Empirical ratio	20	double	1.027	0.9525	1.050	0.9520

All entries use 2,000 planned repetitions. R_V divides mean estimated variance by the sample variance of panel-relative errors. Raw uses observed residuals formed with finite predictions; split adds the exact conditional generation variance to the exact-moment base; omit uses that base alone; double adds generation variance to the raw variance. Omit and double are diagnostic controls. †: the pointwise 95% Wilson interval excludes 0.95. No adjustment is made across conditions. Full Wilson endpoints, MCSEs and interval widths are in the accompanying summary file.

Table D.5 quantifies the small raw-split differences using their paired repetitions. The ratio of mean reported variances ranges from 0.9986 to 1.0028, and raw-minus-split coverage ranges from -0.0010 to 0.0025 . For the heterogeneous-probability empirical high case at $M = 1$, that coverage difference is 0.0025 with paired MCSE 0.0015 . It does not establish that raw repairs the coverage shortfall. Agreement of variance calculations in these settings is also not a finite-sample identity for a fitted learner, as Appendix B explains.

Table D.5: E3: paired differences between raw and split variance calculations.

Law	M	Task	Ratio of mean variances	Coverage difference (MCSE)
<i>E3: constant probabilities</i>				
Gaussian ratio	1	High	0.99878	0.0015 (0.0015)
Gaussian ratio	1	Cap	0.99971	0.0005 (0.0015)
Gaussian ratio	20	High	0.99993	-0.0010 (0.0010)
Gaussian ratio	20	Cap	1.00004	0.0010 (0.0007)
Empirical ratio	1	High	0.99862	-0.0005 (0.0011)
Empirical ratio	1	Cap	0.99961	0.0000 (0.0017)
Empirical ratio	20	High	1.00003	-0.0010 (0.0007)
Empirical ratio	20	Cap	1.00002	0.0010 (0.0007)
<i>E3: heterogeneous probabilities</i>				
Gaussian ratio	1	High	1.00281	0.0020 (0.0017)
Gaussian ratio	1	Cap	1.00106	0.0010 (0.0017)
Gaussian ratio	20	High	0.99993	0.0005 (0.0005)
Gaussian ratio	20	Cap	0.99972	-0.0005 (0.0009)
Empirical ratio	1	High	1.00246	0.0025 (0.0015)
Empirical ratio	1	Cap	1.00078	0.0010 (0.0020)
Empirical ratio	20	High	0.99989	0.0000 (0.0012)
Empirical ratio	20	Cap	0.99973	-0.0005 (0.0005)

Mean variance ratio is mean raw variance divided by mean split variance, not the mean of within-repetition ratios. Coverage difference is raw minus split. Its MCSE uses the paired coverage indicators over 2,000 repetitions. The complete file also reports the mean variance difference and its paired MCSE.

The summaries also retain bias rather than treating it as identically zero. For example, the original SIPP high biases for direct, Gaussian ratio and empirical ratio prediction are respectively 0.000235 , 0.000294 and 0.000272 , with MCSEs 0.000091 , 0.000100 and 0.000095 . These methods share masks, so the positive deviations are not independent findings. The theoretical claims in Appendix B are conditioned or asymptotic statements; the numerical comparisons do not establish exact unbiasedness for the fitted procedures.

D.3 Prediction and threshold diagnostics

Table D.6 separates task predictions from final augmented estimates. For each task, c_h is the fitted task mean and $\bar{c}$ averages these predictions over households and then repetitions. The prediction risk averages $H^{-1} \sum_h (1 - p_h)(q_h - c_h)^2 / p_h$ over repetitions; it uses the complete-panel outcomes only in the evaluator. Its fixed-field relation to estimation variance does not identify it with the MSE of a fitted procedure. These are descriptive averages, without Monte Carlo intervals in this table.

Table D.6: E1 and E2: prediction means and evaluator prediction risks.

Method	Scale	$\bar{c}$		Prediction risk	
		High	Cap	High	Cap
<i>E1: normal residuals</i>					
Direct	All	0.14407	0.45515	0.11009	0.11987
Gaussian ratio	All	0.14058	0.45209	0.10933	0.11932
Empirical ratio	All	0.14071	0.45212	0.10937	0.11932
Independent amounts	0	0.08783	0.40828	0.11334	0.12172
Independent amounts	0.25	0.14059	0.45208	0.10933	0.11931
Independent amounts	1	0.22764	0.54971	0.11664	0.12832
<i>E1: mixture residuals</i>					
Direct	All	0.14944	0.46471	0.12112	0.16531
Gaussian ratio	All	0.14086	0.45240	0.12231	0.16485
Empirical ratio	All	0.14775	0.46170	0.12058	0.16460
Independent amounts	0	0.08815	0.40855	0.12635	0.16799
Independent amounts	0.25	0.14082	0.45233	0.12229	0.16482
Independent amounts	1	0.22777	0.54991	0.12739	0.17183
<i>E2: selected SIPP panel</i>					
Direct	All	0.10139	0.47558	0.08654	0.06021
Gaussian ratio	All	0.21544	0.54649	0.10095	0.06666
Empirical ratio	All	0.13629	0.48568	0.09213	0.06219
Independent amounts	0	0.27109	0.61699	0.11633	0.08103
Independent amounts	0.5	0.29679	0.65386	0.12559	0.09247
Independent amounts	2	0.40039	0.82330	0.17740	0.18205

All denotes the three paired scales in the corresponding experiment. Each entry averages 2,000 repetitions. Prediction means exclude the observed-residual correction and are not final estimates. Prediction risk is a full-panel evaluator diagnostic; hidden-outcome prediction MSEs and fitted residual variances are also retained in `supplementary/prediction_summary.csv`.

On the original SIPP panel, Gaussian ratio prediction has a mean high probability of 0.21544, whereas the complete-panel high share is 0.10117. The augmented estimate can remain much closer to the target because of the residual correction. Reading the prediction mean as estimator bias would therefore give a different, incorrect assessment of the method.

The boundary diagnostic in Table D.7 concerns the fitted empirical predictive law, not the observed frequency of ratios at the threshold. In each fit it averages predictive mass in $[\log(0.8) - \epsilon, \log(0.8) + \epsilon]$ over evaluation households; the table then averages repetitions. E1 and E2 values are identical across scales, so only scale zero is summarized. E3 uses its sole scale.

Table D.7: Empirical predictive mass near the log-ratio threshold.

Experiment	$\epsilon = 0$	10^{-12}	10^{-8}	10^{-4}
E1: normal	0	0	7.634×10^{-9}	5.537×10^{-5}
E1: mixture	0	0	4.207×10^{-9}	4.075×10^{-5}
E2: SIPP panel	2.313×10^{-7}	2.313×10^{-7}	2.387×10^{-7}	7.836×10^{-5}
E3: constant p	0	0	5.443×10^{-9}	5.560×10^{-5}
E3: heterogeneous p	0	0	5.794×10^{-9}	5.587×10^{-5}

Each cell is a mean over 2,000 repetitions. Values are predictive probabilities under the fitted empirical residual law, not proportions of observed raw ratios. The full file retains all scales, repetitions and four window widths.

The selected SIPP panel contains 43 raw ratios equal to 0.8. That count differs from the fitted exact-threshold mass, whose mean is approximately 2.31×10^{-7} . Canonical outcomes continue to use the original ratio comparison, including these boundary cases. Neither the small predictive mass nor the finite set of window widths proves the asymptotic boundary condition required for learner stability.

For the five invariant methods, the recorded per-repetition spans of both estimates and variances across scales are zero. The accompanying `supplementary/scale_invariance.csv` retains all 72,000 family–repetition–method–task records, including nonzero spans for independent amounts. Separate structural records check the Gaussian joint-amount versus ratio calculations and the fitted variance ratio. These identities verify the stated implementation; they do not rank the invariant methods by accuracy.

D.4 Standalone computation times

Tables D.8 and D.9 report all 90 timed configurations. Each entry covers a joint call returning both tasks, rather than separate times for high and cap. All configurations are fitted independently, including invariant methods at each scale. Means and their MCSEs are in milliseconds; the timed boundary and one-thread environment are specified in Appendix C.6.

Table D.8: Standalone computation in E1 and E2: mean milliseconds (MCSE).

Method	Scale zero	Intermediate scale	Largest scale
<i>E1: normal residuals</i>			
HT	1.83 (0.014)	1.80 (0.014)	1.80 (0.012)
Hájek	1.67 (0.010)	1.69 (0.011)	1.68 (0.013)
Direct	39.83 (0.172)	39.86 (0.171)	39.83 (0.179)
Gaussian ratio	9.33 (0.045)	9.33 (0.049)	9.29 (0.048)
Empirical ratio	10.61 (0.049)	10.57 (0.049)	10.61 (0.050)
Independent amounts	10.23 (0.058)	10.10 (0.048)	10.07 (0.051)
<i>E1: mixture residuals</i>			
HT	1.82 (0.013)	1.80 (0.014)	1.82 (0.022)
Hájek	1.72 (0.016)	1.69 (0.016)	1.67 (0.013)
Direct	38.58 (0.193)	38.44 (0.189)	38.79 (0.220)
Gaussian ratio	9.51 (0.053)	9.41 (0.056)	9.45 (0.052)
Empirical ratio	10.80 (0.062)	10.78 (0.064)	10.79 (0.086)
Independent amounts	10.34 (0.063)	10.14 (0.054)	10.21 (0.059)
<i>E2: selected SIPP panel</i>			
HT	3.10 (0.022)	3.02 (0.020)	3.06 (0.022)
Hájek	2.81 (0.020)	2.82 (0.021)	2.78 (0.019)
Direct	67.09 (0.238)	66.94 (0.232)	66.87 (0.234)
Gaussian ratio	16.84 (0.076)	16.57 (0.071)	16.65 (0.074)
Empirical ratio	19.00 (0.080)	19.04 (0.085)	19.02 (0.089)
Independent amounts	17.82 (0.080)	17.73 (0.079)	17.66 (0.076)

Each time summarizes 2,000 standalone calls with no fit cache. Intermediate/largest scales are $s = 0.25, 1$ in E1 and $s = 0.5, 2$ in E2. A call includes canonical-input validation, fitting, predictions, covariance, standard errors, intervals and return status. Raw-file input/output, canonical transformation and evaluator diagnostics are outside the timer. The full timing file also reports medians, standard deviations, minima and maxima.

On the original SIPP panel, the mean time ranges from about 2.81 ms for Hájek to 67.09 ms for direct regression; Gaussian and empirical ratio procedures take about 16.84 and 19.00 ms. Direct regression's lower observed MSE in this panel is therefore accompanied by more computation in this implementation. These are measured costs of the specified fits, not machine-independent complexity bounds.

Table D.9: E3 standalone computation: all variance rules and both probability regimes.

Law	M	Rule	Constant p		Heterogeneous p	
			Mean (MCSE)	Median	Mean (MCSE)	Median
Gaussian ratio	∞	Exact	9.20 (0.047)	8.35	9.99 (0.047)	9.41
Gaussian ratio	1	raw	9.22 (0.067)	8.30	9.91 (0.047)	9.35
Gaussian ratio	1	split	12.84 (0.070)	11.57	13.71 (0.062)	13.02
Gaussian ratio	1	omit	12.71 (0.066)	11.45	13.77 (0.073)	13.10
Gaussian ratio	1	double	12.81 (0.069)	11.61	13.95 (0.077)	13.21
Gaussian ratio	20	raw	11.79 (0.067)	10.68	12.93 (0.078)	12.24
Gaussian ratio	20	split	15.38 (0.082)	13.97	16.74 (0.083)	16.04
Gaussian ratio	20	omit	15.36 (0.085)	13.84	16.64 (0.083)	15.90
Gaussian ratio	20	double	15.40 (0.083)	13.95	16.55 (0.078)	15.75
Empirical ratio	∞	Exact	10.55 (0.054)	9.57	11.38 (0.060)	10.64
Empirical ratio	1	raw	9.81 (0.050)	8.90	10.71 (0.059)	10.08
Empirical ratio	1	split	15.76 (0.088)	14.22	16.98 (0.074)	16.28
Empirical ratio	1	omit	15.76 (0.081)	14.27	16.90 (0.078)	16.17
Empirical ratio	1	double	15.84 (0.082)	14.35	16.97 (0.095)	16.19
Empirical ratio	20	raw	11.94 (0.063)	10.83	13.00 (0.068)	12.30
Empirical ratio	20	split	17.81 (0.094)	16.20	19.37 (0.091)	18.62
Empirical ratio	20	omit	17.94 (0.105)	16.23	19.48 (0.093)	18.72
Empirical ratio	20	double	17.96 (0.098)	16.24	19.44 (0.170)	18.42

All times are milliseconds, with 2,000 complete two-task calls per configuration. $M = \infty$ denotes exact integration. Raw does not require the extra exact task moments used by split, omit and double. The four finite- M rules nevertheless share the same point estimate through the coupled generation stream.

Raw variance computation is faster than split for the recorded finite configurations. Some $M = 1$ raw calls are also slightly faster than exact integration, but their MSE is substantially higher (Table D.3). At $M = 20$, exact integration is both faster and more accurate in these recorded comparisons. The results favor exact moments when available and quantify the separate cost of decomposing generation variance. They do not establish a universal speed advantage for finite generation or for a particular residual law.

D.5 Machine-readable tables

The accompanying `supplementary/` directory contains `summary.csv` (180 performance rows), `paired_comparisons.csv` (328 paired contrasts), `finite_M_summary.csv` (the 72-row E3 subset), `variance_rule_comparisons.csv` (16 raw-split contrasts), `timing_summary.csv` (90 configurations), and `prediction_summary.csv` (36 prediction summaries). The files `scale_invariance.csv`, `structural_checks.csv` and `boundary_diagnostics.csv` retain repetition-level diagnostics; `calibration_review.csv` supplies the pointwise coverage flags. These are copies of the archived confirmation outputs, with source hashes and a field guide. Appendix C and the reproduction notes identify the separate scalar outcome, mask and code records needed to reconstruct them. No additional estimation runs or random draws were used to prepare this appendix.

Data and code availability

The 2023 SIPP public-use data are available from the U.S. Census Bureau at <https://www.census.gov/programs-surveys/sipp/data/datasets/2023-data/2023.html>. The accompanying reproduction materials contain the selected household panel, panel-construction code, frozen experimental inputs, estimation code, archived scalar results and commands to regenerate the reported summaries, tables and figures. The raw public-use ZIP can be obtained from the Census location specified in the reproduction README. The materials distinguish the selected panel from national population estimates and document the original export recovery. No restricted-use Census data are required.

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